S1 Supplementary Material

This supplement documents GaitEval's single evaluation protocol: task-eligible participant partitions, label-redacted pose-derived evidence, declared model and evidence conditions, training-only fitting of supervised transformations, and fixed-task-label scoring with participant-aware interpretation. All six datasets—ASD-Kinect, ATAXIA, CARE-PD, GMFCS, PARKINSONISM, and PATHOGAIT—have equal benchmark membership. Model comparison, representation, adaptation, and transfer are experiment configurations within this protocol. Each result specifies its actual evaluation records and settings. The corrected GMFCS 866-clip participant maps are independently verified; complete training manifests for the seven-condition evaluation remain unavailable. The benchmark does not evaluate diagnosis from complete clinical records.

S1.1 Evaluation Scope, Resources, and Splits

S1.1.1 Evaluation records for model, view, and adaptation experiments

Table S1 lists the larger-test records used for model, input-view, and adaptation experiments. Table S2 lists the seven-condition Qwen-7B records. Within each experiment, methods are compared on matching test IDs, labels, and feature columns. Tests with different eligible records are identified by their counts, not pooled as paired results. ATAXIA and PATHOGAIT concern performed patterns rather than patient-level diagnosis.

Table S1: Records for model, view, and adaptation experiments. Walking clips in training/validation/test partitions. Dataset-specific tasks and participant counts determine interpretation. "–" means that this record configuration is not used for that dataset. Corrected GMFCS has 866 test clips for the verified five-model, SFT, and reference results; matching 7B view results remain unverified. The inconsistent historical 885-clip description is not an evaluation count.

Resource	Population/source	Task	Classes	Train	Val.	Test	Test subjects
GMFCS	cerebral-palsy cohort; corrected split	ordinal severity	I–IV	1,602	478	866	166
ATAXIA	healthy volunteers performing ataxic gait	binary	normal / mimicked ataxia	7,004	1,627	1,647	3
PARKINSONISM	patient cohort	binary severity	mild / severe	420	58	120	5
PATHOGAIT	healthy volunteers performing six patterns	six-way pattern	normal + five abnormal patterns	7,361	1,327	2,181	2
ASD-Kinect	see next table	binary	ASD / typical	–	–	–	–
CARE-PD	see next table	UPDRS gait	0–3	–	–	–	–

Subject identity is the uncertainty-resampling unit. ATAXIA's three model-comparison test participants contribute both normal and imitated-ataxia clips and are clustered across classes. PATHOGAIT has only two test participants, so no population-level superiority claim is made. The original GMFCS partition shared three patients between training and test (14 clips); checking validation identifies additional overlap. The repaired partition has 866 test clips from 166 participants and no participant intersections. The seven-condition GMFCS test has 255 clips from 149 participants. Original-to-corrected index maps preserve the provenance of the split-repair experiment.

S1.1.2 Evaluation records for the seven shared conditions

The seven-condition evaluation uses one Qwen2.5-7B runner, prompt template, parser, and capture-quality-free feature policy across all six datasets (Table S2). CARE-PD uses only the four-level UPDRS gait item: its full corpus has 362 patients and 8,476 clips, while this test subset has 665 clips from 24 patients. ASD-Kinect has a 70/10/20 participant split; the source reports some movements simulated by other people without identifying the affected sequences, so patient-only provenance is not independently established.

Table S2: Records and features for the seven shared conditions. Feature counts reflect supported columns after skeleton-role mapping and removal of capture-quality variables. Test counts are the evaluable records in the released prediction files.

Resource	Train clips	Test clips	Test subjects	Features	Label space
ASD-Kinect	70	20	20	23	ASD / typical
ATAXIA	28	8	4	14	normal / mimicked ataxia
CARE-PD (UPDRS gait)	1,901	665	24	24	scores 0–3
GMFCS	914	255	149	16	levels 1–4
PARKINSONISM	83	29	5	16	mild / severe
PATHOGAIT	5,040	1,440	2	24	six gait patterns

The shared-condition ATAXIA test has eight clips from four participants and supports descriptive conclusions only; the 1,647-clip model/adaptation test is a different recorded population. PARKINSONISM has 29 clips from five participants in the shared conditions and 120 clips from five participants in the model/adaptation experiments. Scores are compared only within their matching test records. CARE-PD score 3 has only 18 test clips, limiting class-specific interpretation.

S1.2 Pose Preprocessing and Model-Visible Input Views

Each walking clip is converted into one model-visible view at a time. Views are never concatenated unless explicitly stated. Answer-revealing labels, sample identifiers, subject identifiers, and split metadata are removed from test-query evidence. Candidate classes define the task; few-shot references deliberately include their training labels.

Table S3: Controlled input representations. GaitEval uses four input-view families. The z-scored named-feature representation is a derived scaling control within the named-feature family. Motion summaries and feature views are deterministic functions of the pose sequence.

View	Shape/format	Construction	Evaluation role
Raw coordinates	$124 \times J_d \times 3$	valid pose frames, zero-padded to fixed length	numeric-sequence stress test
Normalized coordinates	$32 \times 12 \times 3$	common joints, body-centered, scale/direction normalized	cross-layout coordinate view
Motion summary	fixed-section English text	deterministic verbalization of measured features	primary readable evidence view
Raw feature vector	named values	supported biomechanical measurements	exact-feature evidence test
Z-scored named feature vector	named standardized values	training-mean/SD normalization	scale-controlled evidence test

S1.2.1 Raw and normalized coordinate views

Raw-coordinate experiment clips use a fixed tensor of 124 frames by the dataset-specific joint count J_d and three channels. Shorter clips are zero-padded; feature and summary computation uses only valid frames. At the nominal 30 frames/s, 124 frames correspond to approximately 4.1 seconds. Temporal measurements are expressed per frame or as dimensionless ratios so that no label-dependent temporal normalization is introduced.

The normalized coordinate view maps each skeleton to 12 common joints in the fixed order shoulders, elbows, wrists, hips, knees, and ankles. For every frame, the midpoint of the hips is subtracted from all joints. Coordinates are divided by the mean shoulder-center–to–hip-center distance computed over valid frames. The principal direction of the hip-center trajectory is aligned with the positive x axis, and the remaining axes are defined by an orthogonal body-centered basis. Thirty-two frames are then selected at uniform temporal indices. The prompt contains a $32 \times 12 \times 3$ table rounded to two decimals. This view preserves relative motion while reducing differences in global location, scale, orientation, joint count, and clip duration.

S1.2.2 Deterministic biomechanical features

For joint j, spatial channel c, and valid frame set T, range of motion is

$$R_{j,c} = \max_{t \in T} p_{t,j,c} - \min_{t \in T} p_{t,j,c}. \tag{1}$$

Confidence channels are never used as spatial coordinates. For paired nonnegative measurements v_L and v_R, unsigned asymmetry is

$$A(v_L, v_R) = \frac{|v_L - v_R|}{0.5(|v_L| + |v_R|) + 10^{-8}}. \tag{2}$$

The ratio lies in $[0, 2]$ but discards the direction of asymmetry; this is a known limitation for unilateral presentations. The jerk proxy is the mean absolute third finite difference of the relevant trajectory. Trajectory variability is the temporal variance of the specified joint path. The standardized value is

$$z_{i,f} = \frac{x_{i,f} - \mu_f^{\text{train}}}{\sigma_f^{\text{train}} + 10^{-8}}, \tag{3}$$

using training-split statistics only.

Table S4: Complete deterministic feature inventory. Braces denote separate left and right features. Skeleton layouts that do not support a semantic role leave the corresponding feature undefined.

Group	Released feature name	Definition
Amplitude	`ankle_{L,R}_vert_range`	vertical ankle range
Amplitude	`ankle_{L,R}_horiz_range`	horizontal ankle range
Amplitude	`knee_{L,R}_vert_range`	vertical knee range
Amplitude	`hip_{L,R}_vert_range`	vertical hip range
Asymmetry	`ankle_vert_asymmetry`	paired ankle vertical-range asymmetry
Asymmetry	`ankle_horiz_asymmetry`	paired ankle horizontal-range asymmetry
Asymmetry	`knee_vert_asymmetry`	paired knee vertical-range asymmetry
Asymmetry	`hip_vert_asymmetry`	paired hip vertical-range asymmetry
Stability	`trunk_lateral_sway`	lateral trunk range
Stability	`trunk_vert_range`	vertical trunk range
Stability	`hip_centre_lateral_std`	lateral hip-center standard deviation
Stability	`hip_centre_vert_std`	vertical hip-center standard deviation
Variability	`mean_joint_trajectory_var`	mean joint-path variance
Variability	`ankle_{L,R}_traj_var`	left/right ankle-path variance
Smoothness	`ankle_{L,R}_jerk`	left/right ankle jerk proxy
Smoothness	`ankle_mean_jerk`	mean of the ankle jerk proxies
Smoothness	`trunk_jerk`	trunk jerk proxy
Capture audit	`valid_frame_fraction`	fraction of valid frames in the fixed tensor

The inventory above describes the coordinate-derived extractor, with 23 motion measurements and the capture-quality field `valid_frame_fraction`. Supported feature columns depend on skeleton layout. The seven shared conditions use 14/16/16/24/23/24 features for ATAXIA/GMFCS/PARKINSONISM/PATHOGAIT/ASD-Kinect/CARE-PD and exclude capture-quality variables, including clip length and valid-frame fraction. The larger-test model/view/adaptation inputs retain their recorded supported columns; the explicit PATHOGAIT field-removal experiment quantifies its effect on matching IDs. Missing joints are omitted from both LLM and reference inputs. Reference models impute supported missing values with training medians; text prompts omit undefined values. Complete feature CSVs remain external, so identical columns across different experiments are not assumed.

S1.2.3 Motion-summary provenance and template

Motion summaries are deterministic, label-independent renderings of the selected feature vector, not annotations or outputs of another language model. Sections follow MOTION AMPLITUDE, SYMMETRY/ASYMMETRY, STABILITY, TRAJECTORY VARIABILITY, and SMOOTHNESS. The larger-test summary generator additionally emits a CLIP QUALITY line. This line and its underlying feature are excluded from all seven shared conditions; the feature-removal experiment separately evaluates the 2,181-clip PATHOGAIT record set.

Raw amplitudes and ranges are printed to four decimals; variances and jerk proxies are printed to six decimals. A small set of fixed thresholds adds qualitative wording without using a label:

- asymmetry < 0.05: symmetric; $[0.05, 0.15)$: mild; $[0.15, 0.30)$: notable; ≥ 0.30: strong;
- trunk lateral sway < 0.04: mild; $[0.04, 0.09)$: moderate; ≥ 0.09: marked;
- ankle mean jerk $< 2 \times 10^{-4}$: smooth; $[2 \times 10^{-4}, 8 \times 10^{-4})$: moderately smooth; $\geq 8 \times 10^{-4}$: jerky.

The thresholds only select descriptive phrases; the underlying numerical values are retained. A typical summary is 150–300 words (approximately 320 tokens). The following schematic preserves the exact structure while using illustrative values:

```
MOTION AMPLITUDE
Left/right ankle vertical range:  0.0900 / 0.1400.
Left/right knee vertical range:  0.0710 / 0.0840.

SYMMETRY/ASYMMETRY
Ankle vertical asymmetry:  0.4350 (strong asymmetry).

STABILITY
Trunk lateral sway:  0.0400 (moderate sway).
```

```
TRAJECTORY VARIABILITY
Mean joint trajectory variance:  0.008000.

SMOOTHNESS
Mean ankle jerk:  0.000700 (moderately smooth).
```

The candidate label set is inserted only by the downstream classification prompt. It is never available to the summary generator, so the motion evidence and task instruction remain separate objects.

S1.3 Exact Prompt Formats and Output Parsing

S1.3.1 Flat zero-shot prompt

The shared-condition flat-prompt template is instantiated as follows (line wrapping is for display), where bracketed fields are filled with the selected evidence view and the dataset's complete candidate list:

```
You are classifying a gait clip from a compact biomechanical
[EVIDENCE_KIND] derived from a skeleton sequence.

[EVIDENCE]

Task: based ONLY on the [EVIDENCE_KIND] above, classify the gait
into exactly one of the following classes:
- [CLASS 1]
- [CLASS 2]
...

Return only one label, written exactly as listed above.
```

For motion summaries, `EVIDENCE_KIND` is "motion summary"; for raw or z-scored features it is "feature description." The larger-test view experiments use the recorded 3D-skeleton/biomechanical-dimension preamble. These template settings are documented with the experiment, and do not supply clinical definitions, prevalence, demonstrations, or answer labels in zero-shot queries.

S1.3.2 Reasoning-anchored prompt

The reasoning-anchored condition used the same label-redacted motion summary as the flat condition, but replaced the direct-answer instruction with the following structured prompt. No separate system message was used; the complete prompt was sent as a single user message. Bracketed fields below denote deterministic run-time substitutions.

```
You are a gait-analysis expert classifying a walking clip from a compact biomechanical motion summary derived from a 3D
↪  skeleton sequence. The summary reports motion amplitudes, left/right asymmetry, trunk/pelvis stability, trajectory
↪  variability, and movement smoothness (jerk).

[MOTION_SUMMARY]

Candidate classes:
[CANDIDATE_CLASS_LINES]

Reason step by step:
1. List the 2-3 most abnormal biomechanical signals in this summary (e.g. high asymmetry, high jerk, large trunk sway,
↪  reduced amplitude).
2. For EACH candidate class, briefly state whether those signals are consistent or inconsistent with it.
3. Do NOT default to the most common class; pick the class best supported by the specific signals.
Finally, on a new line, output exactly: FINAL: <one class name>
```

The candidate lines were inserted in ascending numeric-label order using the exact dataset label strings. This condition used Qwen2.5-72B-Instruct with temperature 0 and a maximum of 512 output tokens. For ATAXIA and PATHOGAIT, the test records were shuffled with seed 0 before selecting 150 clips. PARKINSONISM contains only 120 test clips, so all of its test clips were evaluated after the same seeded shuffle. The generated reasoning text was not treated as explanation supervision or scored for clinical correctness; only the parsed final class was evaluated.

For the three evaluated resources, the exact candidate strings were:

- ATAXIA: `normal gait`; `cerebellar ataxic gait`.
- PARKINSONISM: `low motor severity`; `high motor severity`.
- PATHOGAIT: `normal gait`; `antalgic gait`; `lurching gait`; `steppage gait`; `stiff-legged gait`; and `Trendelenburg gait`.

S1.3.3 Few-shot prompt

Few-shot prompting was implemented as one user message containing a sequence of labeled textual reference blocks, followed by the unlabeled test query. It was not implemented as a multi-turn sequence of separate user and assistant messages. No separate system message was used. The literal template was:

```
You are classifying a gait clip from a compact biomechanical motion summary derived from a 3D skeleton sequence.
↪  Feature values (amplitudes, left/right asymmetry, stability, trajectory variability, smoothness) must be judged
↪  RELATIVE TO THIS DATASET, not by generic clinical intuition. Below are [N_SHOTS] labeled reference clips from the
↪  same dataset and feature pipeline; use them to calibrate what each class looks like here.

=== Reference example 1 ===
[REFERENCE_MOTION_SUMMARY_1]
Ground-truth class: [REFERENCE_CLASS_1]

=== Reference example 2 ===
[REFERENCE_MOTION_SUMMARY_2]
Ground-truth class: [REFERENCE_CLASS_2]

[ADDITIONAL_REFERENCE_BLOCKS]

=== Now classify this new clip ===
[QUERY_MOTION_SUMMARY]

Candidate classes:
[CANDIDATE_CLASS_LINES]

Return only one label, written exactly as listed above.
```

Reference examples were drawn only from the dataset's training split. For a requested context size k and C classes, the implementation grouped training examples by numeric class, shuffled each class pool with seed 0, selected $\max(1, \lfloor k/C \rfloor)$ examples from each class, and then shuffled the combined reference set using the same random-number generator. All reported context sizes were divisible by the corresponding number of classes, so the realized reference set contained exactly k examples and was class-balanced. One fixed reference set was reused for every test clip within a run.

Inference used Qwen2.5-72B-Instruct, temperature 0, a maximum of 64 output tokens, and the same exact-label parser used for the corresponding flat-prompt experiments.

The reported settings were:

- ATAXIA: $k \in \{8, 16, 32\}$, corresponding to 4, 8, and 16 references per class, evaluated on the full test split.
- PARKINSONISM: $k = 16$, corresponding to 8 references per class, evaluated on the full test split.
- PATHOGAIT: $k = 12$, corresponding to 2 references per class. This run evaluated the first 600 records in the stored test-file order rather than the complete 2,181-clip test split.

S1.3.4 Parser and invalid outputs

The reasoning condition did not use a dedicated parser for the `FINAL:` field. The reasoning and few-shot outputs were parsed with the same exact-label parser as the corresponding flat-prompt runs. Outputs were whitespace-normalized and lowercased, and candidate strings were matched with leading and trailing alphanumeric boundaries. When several complete candidate strings occurred, the latest occurrence was selected; positional ties favored the longer candidate string. Consequently, a compliant final `FINAL: <class>` line was normally selected because it was the last candidate mention, rather than because the parser explicitly recognized the `FINAL:` prefix.

If no complete candidate occurred, the larger-test view/few-shot implementation additionally accepted an output-ending substring only when it was a unique prefix of exactly one candidate label, providing a limited fallback for truncated generations. All other outputs were marked `UNPARSED`. Unparsed outputs were never discarded: they were counted as errors and contributed a false negative to the true class.

S1.4 Models, Inference, and Adaptation

S1.4.1 Zero-shot model roster

The five-model motion-summary experiment uses Llama-3.3-70B-Instruct, Qwen2.5-72B-Instruct, Gemma-3-27B-IT, GLM-4-32B-0414, and DeepSeek-R1-Distill-Qwen-32B. All use deterministic decoding through vLLM. Test clips are 20, 1,647, 665, 866, 120, and 2,181 for ASD-Kinect, ATAXIA, CARE-PD, GMFCS, PARKINSONISM, and PATHOGAIT. The shared-condition evaluation instead uses Qwen2.5-7B on 20/8/665/255/29/1,440 clips, as listed in the evaluation-record tables.

Generation limits are recorded per experiment: the shared-condition Qwen-7B runs and ASD-Kinect/CARE-PD non-reasoning model runs use 48 tokens; the larger-test view launcher uses 16; the other direct-answer model and few-shot runs use 64. Reasoning-distilled runs use 2,048 tokens. All use temperature 0. Few-shot and reasoning-anchored probes use Qwen2.5-72B. The larger-test parser has a unique truncated-prefix fallback, while the seven-condition parser requires complete candidate mentions. These settings preclude matched-budget or identical-parser claims across different experiments. Within each summary-SFT comparison, its zero-shot and SFT prompts are byte-identical.

S1.4.2 LoRA supervised fine-tuning

For the seven shared conditions, LoRA-SFT is trained separately for each dataset on training-split motion summaries and their candidate labels. The configuration is rank $r = 16$, scaling $\alpha = 32$, dropout 0.05, and adapters on all linear projections. Training uses completion-only cross-entropy, three epochs, learning rate 10^{-4} with cosine decay, maximum sequence length 1,024, per-device batch size 8, and gradient accumulation 2 (effective batch size 16). The committed trainer reads only training examples, uses no evaluation dataset and no intermediate checkpoint selection (`save_strategy="no"`), and saves the final three-epoch adapter. A validation-selected checkpoint is not supported by this implementation. The configured test split is not used for training or normalization.

The shared-condition training-set sizes are 28 for ATAXIA, 914 for GMFCS, 83 for PARKINSONISM, 5,040 for PATHOGAIT, 70 for ASD-Kinect, and 1,901 for CARE-PD. CARE-PD training examples are restricted to the UPDRS gait task. SFT therefore measures recovery of an in-distribution, dataset-specific mapping from supplied evidence to the dataset label space; it is not interpreted as acquisition of a general diagnostic rule.

S1.4.3 Few-shot sweep

Table S5: Few-shot motion-summary prompting. Qwen2.5-72B receives balanced labeled training examples. Cells report accuracy / macro-F1 / Cohen's κ. Bold marks the best tested context size within a dataset.

Dataset	Setting	Accuracy / macro-F1 / κ
ATAXIA	zero-shot	0.685 / 0.406 / -0.002
ATAXIA	$k = 8$	0.682 / 0.644 / 0.289
ATAXIA	$k = 16$	**0.740 / 0.697 / 0.395**
ATAXIA	$k = 32$	0.642 / 0.633 / 0.299
PARKINSONISM	zero-shot	0.575 / 0.472 / 0.035
PARKINSONISM	$k = 16$	**0.567 / 0.561 / 0.188**
PATHOGAIT	zero-shot	0.155 / 0.055 / -0.009
PATHOGAIT	$k = 12$	**0.268 / 0.185 / 0.081**

The ATAXIA curve is non-monotonic, so few-shot prompting is treated as an adaptation probe rather than a tuned final model. PATHOGAIT uses two references per class at $k = 12$, but its 600-clip few-shot subset differs from the 2,181-clip zero-shot row; their difference is not a paired full-test gain.

S1.5 Matched Linear References

The supervised references are multinomial or binary logistic regression and a linear support-vector machine. Both receive exactly the supported deterministic feature columns available to the feature-view LLM condition. The supervised references use prespecified estimator settings: logistic regression is fitted with `max_iter=2000`, and the linear SVM with `max_iter=5000`. The class-balanced configuration is the primary reference, while the unweighted configuration is reported as a parallel sensitivity analysis rather than selected on validation performance. Missing values are imputed using training-set medians, and the imputer

and standard scaler are fitted on the training split only before being applied to the held-out test split. No validation-based hyperparameter search is performed. Columns missing for an entire dataset are dropped, and residual missing values are imputed with the training median. These models are signal probes: they establish whether the deterministic pose representation contains label-relevant information under supervised learning, rather than representing the state of the art in clinical gait analysis.

Table S6: Feature-reference weighting check on the larger-test records. "–" denotes no result. Macro-F1 on the specified model/adaptation test sets. "Balanced" uses inverse-frequency class weights; "none" uses the empirical training distribution.

Reference	ATAXIA	GMFCS	PARKINSONISM	PATHOGAIT
Logistic regression (balanced)	0.763	0.395	0.613	0.662
Logistic regression (none)	0.773	0.418	0.627	0.663
Linear SVM (balanced)	0.759	0.412	0.583	0.644
Linear SVM (none)	0.768	0.400	0.579	0.646
ASD-Kinect	–	–	–	–
CARE-PD	–	–	–	–

On the 2,181-clip PATHOGAIT test, removing `valid_frame_fraction` reduces balanced LR macro-F1 from 0.662 to 0.564 and linear-SVM macro-F1 from 0.644 to 0.560. A pre-repair sensitivity check recorded changes below 0.001 for the other tested resources, but does not establish that bound for corrected GMFCS. The feature-SFT results are therefore interpreted using their supplied measurements, without a kinematics-only claim. The seven shared conditions exclude capture-quality variables for all methods.

S1.6 Metrics and Statistical Analysis

Let C_{ab} denote the number of clips whose true class is a and predicted class is b, for a fixed candidate set of K classes. Let u_a count invalid predictions for true class a, so $N = \sum_{a,b=1}^{K} C_{ab} + \sum_{a=1}^{K} u_a$. For class k, precision, recall, and F1 are

$$P_k = \frac{C_{kk}}{\sum_{a=1}^{K} C_{ak}}, \qquad R_k = \frac{C_{kk}}{\sum_{b=1}^{K} C_{kb} + u_k}, \tag{4}$$

and

$$F_{1,k} = \frac{2P_k R_k}{P_k + R_k}. \tag{5}$$

A metric is set to zero when its denominator is zero. Macro-F1 is the unweighted mean over the complete dataset-defined class set,

$$\mathrm{MacroF1} = \frac{1}{K} \sum_{k=1}^{K} F_{1,k}, \tag{6}$$

including classes that receive no predictions. Accuracy is

$$\mathrm{Accuracy} = \frac{1}{N} \sum_{k=1}^{K} C_{kk}. \tag{7}$$

These metrics are reported together because accuracy alone can obscure class-allocation failures on imbalanced tasks [16, 17].

For binary tasks, sensitivity is recall for the dataset-specified positive class, whereas specificity is recall for the complementary negative class. Balanced accuracy is

$$\mathrm{BalancedAccuracy} = \frac{\mathrm{Sensitivity} + \mathrm{Specificity}}{2}. \tag{8}$$

Cohen's agreement coefficient is

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \qquad p_o = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(y_i = \hat{y}_i), \tag{9}$$

where y_i and $\hat{y}_i$ are the true and parsed predicted labels for clip i. Chance agreement is computed from the empirical target and prediction marginals:

$$p_e = \sum_{k=1}^{K} \left(\frac{n_k^{(y)}}{N} \right) \left(\frac{n_k^{(\hat{y})}}{N} \right), \tag{10}$$

where $n_k^{(y)}$ and $n_k^{(\hat{y})}$ are the numbers of true and predicted instances of class k, respectively [65]. An unparsed output is assigned a prediction-only sentinel category outside the candidate set. It is therefore always incorrect, contributes a false negative to its true class, and cannot create a true positive for any candidate class.

For ordinal GMFCS and CARE-PD tasks, mean absolute error (MAE) is computed over the ordered class indices:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} \left| r(y_i) - r(\hat{y}_i) \right|, \tag{11}$$

where $r(\cdot)$ maps a class label to its ordinal level. We additionally report quadratic-weighted Cohen's κ. Let

$$w_{ab} = \frac{(a-b)^2}{(K-1)^2} \tag{12}$$

be the quadratic disagreement weight, $O_{ab} = C_{ab}$ the observed confusion matrix, and

$$E_{ab} = \frac{n_a^{(y)} n_b^{(\hat{y})}}{N} \tag{13}$$

the expected count under independent target and prediction marginals. Quadratic-weighted agreement is

$$\kappa_{\mathrm{q}} = 1 - \frac{\sum_{a=1}^{K}\sum_{b=1}^{K} w_{ab} O_{ab}}{\sum_{a=1}^{K}\sum_{b=1}^{K} w_{ab} E_{ab}}. \tag{14}$$

This metric gives larger penalties to errors spanning more ordinal levels. The three plotted ordinal conditions have no invalid outputs; MAE and weighted agreement are reported only for those valid ordered-label cells. Per-class metrics and confusion matrices cover the three plotted conditions, rather than every control condition.

Uncertainty is primarily quantified using a participant-cluster bootstrap with $B = 2{,}000$ draws. For each draw, the test participants are sampled with replacement, and all clips from each sampled participant are retained together. The metric is recomputed on every valid draw (at least two distinct participants and two true classes), and the 2.5th and 97.5th percentiles form the 95% confidence interval [66]. For a paired comparison, the same sampled participants and matched clips are used for both conditions. If

$$\Delta = T_A - T_B \tag{15}$$

is the observed macro-F1 difference, the paired confidence interval is obtained from the bootstrap distribution of Δ^*. The reported two-sided paired p value is estimated from the proportions of participant-bootstrap differences on the two sides of zero. These paired tail areas summarize changes while preserving within-participant dependence. Point metrics here retain the fixed task label set. The archived bootstrap code instead uses the union of observed true/predicted labels, including an invalid-output sentinel and omitting absent classes. Thus, its intervals and tail areas are historical diagnostics, not independently regenerated fixed-label uncertainty estimates.

The per-cell permutation-against-chance values in the shared-condition grid use a separate row-level label-permutation test. The true labels are permuted relative to the fixed predictions while preserving their empirical marginal counts. With observed statistic T_{obs} and permuted statistics $T^{(b)}$, the one-sided Monte Carlo value is

$$p_{\mathrm{perm}} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left(T^{(b)} \geq T_{\mathrm{obs}}\right)}{B+1}. \tag{16}$$

With $B = 2{,}000$, the minimum attainable value is approximately 0.0005. Because this permutation operates on individual records rather than participant clusters, it is treated as a descriptive null comparison and not as participant-level population inference. The archived cluster intervals and paired tail areas retain their stated historical metric convention. The shared-condition permutation calculations use 2,000 draws; reference-table values from the larger-test experiments are retained at their source precision rather than assigned the same Monte Carlo budget.

The limited numbers of independent participants constrain these analyses. The larger-test ATAXIA and PARKINSONISM test sets contain three and five participants, respectively. PATHOGAIT contains only two test participants, for which participant-cluster resampling is degenerate; its frozen-split results are therefore reported as descriptive point estimates with participant-specific values and are not used to support population-level superiority. ASD-Kinect contains one clip per participant, so record-level and participant-level resampling coincide. The highest GMFCS and CARE-PD ordinal levels also have small supports and should be interpreted jointly with their per-class results and confusion matrices.

S1.7 Five-Model Zero-Shot Results

Table S7 gives the full five-model motion-summary results on the test sets listed in its caption. Across the 1,647/866/120/2,181-clip ATAXIA/GMFCS/PARKINSONISM/PATHOGAIT tests, κ ranges from -0.069 to 0.043 after the GMFCS repair. CARE-PD and ASD-Kinect have their own reported ranges. Dominant-class shares show that high accuracy can coexist with concentrated outputs. PATHOGAIT Llama/GLM macro-F1 is .062/.069 over six task labels; earlier .053/.059 values averaged an extra invalid-output category. Invalid responses continue to count as errors.

Table S7: Five-model zero-shot motion-summary results. Accuracy / macro-F1 / Cohen's κ; top share is the dominant predicted-class fraction. Test clips in dataset order ASD-Kinect, ATAXIA, CARE-PD, GMFCS, PARKINSONISM, and PATHOGAIT are 20/1,647/665/866/120/2,181. GMFCS uses the corrected test participants. These are model-comparison records; the seven-condition evaluation's counts are specified separately in the evaluation-record inventory.

Dataset	Model	Accuracy	Macro-F1	κ	Top share
ATAXIA	Llama-3.3-70B	0.686	0.407	0.000	1.00
	Qwen2.5-72B	0.685	0.406	-0.002	1.00
	Gemma-3-27B	0.686	0.407	0.000	1.00
	GLM-4-32B	0.686	0.407	0.000	1.00
	DeepSeek-R1-Distill-Qwen-32B	0.684	0.408	-0.002	1.00
GMFCS	Llama-3.3-70B	0.353	0.212	0.025	0.72
	Qwen2.5-72B	0.372	0.222	0.043	0.78
	Gemma-3-27B	0.313	0.163	0.030	0.86
	GLM-4-32B	0.269	0.127	-0.000	0.93
	DeepSeek-R1-Distill-Qwen-32B	0.311	0.191	0.014	0.64
PARKINSONISM	Llama-3.3-70B	0.583	0.368	0.000	1.00
	Qwen2.5-72B	0.575	0.472	0.035	0.86
	Gemma-3-27B	0.583	0.368	0.000	1.00
	GLM-4-32B	0.583	0.368	0.000	1.00
	DeepSeek-R1-Distill-Qwen-32B	0.583	0.368	0.000	1.00
PATHOGAIT	Llama-3.3-70B	0.159	0.062	-0.010	0.94
	Qwen2.5-72B	0.155	0.055	-0.009	0.94
	Gemma-3-27B	0.165	0.047	-0.001	1.00
	GLM-4-32B	0.141	0.069	-0.069	0.51
	DeepSeek-R1-Distill-Qwen-32B	0.166	0.094	-0.007	0.48
ASD-Kinect	Llama-3.3-70B	0.500	0.451	0.000	0.80
	Qwen2.5-72B	0.450	0.373	-0.100	0.85
	Gemma-3-27B	0.450	0.310	-0.100	0.95
	GLM-4-32B	0.450	0.373	-0.100	0.85
	DeepSeek-R1-Distill-Qwen-32B	0.600	0.583	0.200	0.70
CARE-PD	Llama-3.3-70B	0.441	0.262	0.119	0.71
	Qwen2.5-72B	0.445	0.268	0.129	0.64
	Gemma-3-27B	0.439	0.249	0.113	0.72
	GLM-4-32B	0.441	0.350	0.124	0.56
	DeepSeek-R1-Distill-Qwen-32B	0.398	0.295	0.099	0.45

These results bound the primary claim: the evaluated off-the-shelf models do not consistently ground dataset labels in the supplied motion summaries. They do not imply that pose contains no signal or that every possible language-model representation must fail.

S1.8 Representation and Prompt Controls

S1.8.1 Same-model view comparison

Table S8 reports the Qwen2.5-7B view experiment on 20/1,647/665/866/120/2,181 test clips in dataset order. Red GMFCS entries are displayed for author review only: the summary value comes from a logged 866-clip cross-stack run, normalized coordinates from an unverified source draft, and both feature values from 899-clip aggregates. They do not establish a matched-test comparison and are excluded from scientific comparisons. No view is consistently successful across the tested populations. In the seven-condition evaluation, PARKINSONISM z-scored feature view reaches 0.757 macro-F1 on 29 clips from five participants (Table S9). This result indicates that evidence use can depend on both the dataset and the input representation, although the small test cohort precludes a broad population-level conclusion.

Table S8: Qwen2.5-7B zero-shot input-view experiment. Accuracy / macro-F1 / κ on 20/1,647/665/866/120/2,181 clips in dataset order ASD-Kinect, ATAXIA, CARE-PD, GMFCS, PARKINSONISM, and PATHOGAIT. "–" denotes no recoverable numeric result. Red denotes additions for author review. GMFCS summary is log-only (866 clips); normalized coordinates are draft-only/unverified; raw/z-scored features use 899 clips. These are not verified matched-test results. PATHOGAIT raw features use a recovered 2,181-clip recompute aggregate. The record audit is below. The seven-condition test sets have their own input-view table.

Dataset	Motion summary	Normalized coordinates	Raw feature view	Z-scored feature view
ATAXIA	0.692 / 0.452 / 0.046	0.316 / 0.244 / 0.002	0.382 / 0.347 / 0.057	0.583 / 0.495 / −0.008
GMFCS	0.259 / 0.113 / −0.009	0.282 / 0.110 / 0.000	0.343 / 0.245 / 0.007	0.185 / 0.100 / −0.049
PARKINSONISM	0.550 / 0.400 / −0.045	0.442 / 0.441 / −0.095	0.417 / 0.294 / 0.000	0.508 / 0.493 / −0.014
PATHOGAIT	0.142 / 0.087 / −0.024	0.110 / 0.034 / 0.000	0.110 / 0.033 / 0.000	0.150 / 0.130 / −0.011
ASD-Kinect	0.450 / 0.310 / -0.100	–	0.500 / 0.333 / 0.000	0.550 / 0.436 / 0.100
CARE-PD	0.433 / 0.321 / 0.152	–	0.332 / 0.221 / 0.039	0.233 / 0.133 / 0.020

A separate raw-coordinate stress test supplies Llama-3.3-70B with the $124 \times J_d \times 3$ sequence directly. Scoring invalid outputs as errors gives ATAXIA accuracy/macro-F1/κ of 0.573/0.323/0.009 with 6.0% unparsed outputs, and PATHOGAIT 0.093/0.083/0.038 with 59.3% unparsed outputs. Direct coordinate access therefore does not establish reliable zero-shot classification and creates a substantial formatting burden on the longer six-class task.

S1.8.2 Reasoning-anchored prompt

Qwen2.5-72B is also tested with the explicit compare-then-answer prompt. On fixed 150-clip subsets of ATAXIA and PATHOGAIT and the complete 120-clip PARKINSONISM test set, the reasoning-anchored results are 0.687/0.427/0.028, 0.167/0.105/−0.018, and 0.583/0.368/0.000, respectively. These cells are not clip-matched to the full-split flat-prompt table and are treated as prompt-format diagnostics. The prompt changes some output distributions but does not establish beyond-chance agreement.

Table S9: Complete shared-condition grid (source-reported splits). All cells use Qwen2.5-7B-Instruct. Point macro-F1 is recomputed with the fixed task labels. Intervals and row-level permutation p values are archived diagnostics under the historical label-union convention described in Sec. S1.6; permutation values are not participant-level tests. ‡PATHOGAIT contains only two test participants; its cells are descriptive point estimates, and the degenerate bootstrap intervals and row-level permutation p values are therefore omitted. The CARE-PD 12-class mixed-label cell combines different label tasks and is retained only as a diagnostic, not as a benchmark result.

Resource	Condition	n	Macro-F1	95% participant CI	Permutation p
CARE-PD	Summary	665	0.321	[0.262, 0.404]	0.0005
	Z-features	665	0.133	[0.086, 0.184]	0.68
	Labels reversed	665	0.252	[0.133, 0.293]	0.0005
	Labels shuffled	665	0.285	[0.217, 0.353]	0.0005
	Shuffled evidence	665	0.256	[0.217, 0.313]	0.23
	No evidence	665	0.133	[0.083, 0.183]	1.0
	LoRA-SFT summary	665	0.445	[0.260, 0.507]	0.0005
GMFCS	Summary	255	0.265	[0.217, 0.308]	0.042
	Z-features	255	0.150	[0.108, 0.197]	0.42
	Labels reversed	255	0.322	[0.271, 0.369]	0.006
	Labels shuffled	255	0.274	[0.225, 0.320]	0.13
	Shuffled evidence	255	0.212	[0.172, 0.252]	0.62
	No evidence	255	0.000	[0.000, 0.000]	1.0
	LoRA-SFT summary	255	0.467	[0.413, 0.624]	0.0005
PARKINSONISM	Summary	29	0.396	[0.222, 0.530]	0.52
	Z-features	29	0.757	[0.543, 0.906]	0.0075
	Labels reversed	29	0.356	[0.136, 0.617]	0.90
	Labels shuffled	29	0.356	[0.136, 0.617]	0.90
	Shuffled evidence	29	0.310	[0.104, 0.439]	1.0
	No evidence	29	0.326	[0.133, 0.462]	1.0
	LoRA-SFT summary	29	0.542	[0.255, 0.720]	0.29
ATAXIA	Summary	8	0.273	[0.111, 0.333]	1.0
	Z-features	8	0.333	[0.333, 0.333]	1.0
	Labels reversed	8	0.273	[0.111, 0.333]	1.0
	Labels shuffled	8	0.273	[0.111, 0.333]	1.0
	Shuffled evidence	8	0.273	[0.111, 0.333]	1.0
	No evidence	8	0.333	[0.333, 0.333]	1.0
	LoRA-SFT summary	8	0.333	[0.333, 0.333]	1.0
PATHOGAIT‡	Summary	1,440	0.069	–	–
	Z-features	1,440	0.139	–	–
	Labels reversed	1,440	0.121	–	–
	Labels shuffled	1,440	0.093	–	–
	Shuffled evidence	1,440	0.071	–	–
	No evidence	1,440	0.048	–	–
	LoRA-SFT summary	1,440	0.451	–	–
ASD-Kinect	Summary	20	0.310	[0.200, 0.394]	1.0
	Z-features	20	0.436	[0.259, 0.642]	0.50
	Labels reversed	20	0.341	[0.167, 0.524]	0.96
	Labels shuffled	20	0.341	[0.167, 0.524]	0.96
	Shuffled evidence	20	0.436	[0.259, 0.642]	0.51
	No evidence	20	0.333	[0.231, 0.412]	1.0
	LoRA-SFT summary	20	0.310	[0.200, 0.394]	1.0
CARE-PD	12-class mixed diagnostic	970	0.004	[0.001, 0.009]	1.0

The seven-condition table has two important qualifications. First, PARKINSONISM z-features reach 0.757 macro-F1 on only 29 clips from five subjects; the subject-cluster interval is 0.543–0.906 and overlaps the matched logistic-reference interval of 0.416–0.654. This is evidence that explicit standardized features can be usable by the model in a particular condition, not a population-level superiority result. Second, the ATAXIA shared-condition split contains eight clips and has insufficient power for model ranking.

S1.8.3 Evidence and label-order controls

Table S10: Complete shared-condition paired contrasts. Each contrast is the macro-F1 of the first named condition minus that of the second, computed from unrounded fixed-label scores. Parenthesized values are archived two-sided participant-bootstrap tail areas under the historical label-union convention in Sec. S1.6; they are not newly recomputed fixed-label tests. Thus, a positive value favors the first condition and a negative value favors the second; for example, negative Summary−ZFeat favors Z-features, whereas positive SFT−Summary favors LoRA-SFT. A signed point difference alone is not interpreted as evidence of an effect without its paired uncertainty. ‡PATHOGAIT entries are point differences only because two participants do not support reliable participant-level inference.

Resource	Summary−NoEv	Summary−ShufEv	Summary−ZFeat	Summary−Rev.	Summary−Shuf.	SFT−Summary
CARE-PD	$+0.188$ (0.001)	$+0.065$ (0.029)	$+0.187$ (0.002)	$+0.068$ (0.028)	$+0.036$ (0.045)	$+0.124$ (0.43)
GMFCS	$+0.265$ (0.001)	$+0.052$ (0.089)	$+0.115$ (0.004)	-0.057 (0.030)	-0.010 (0.52)	$+0.202$ (0.001)
PARKINSONISM	$+0.070$ (0.62)	$+0.086$ (0.14)	-0.362 (0.001)	$+0.040$ (0.62)	$+0.040$ (0.62)	$+0.146$ (0.47)
ATAXIA	-0.061 (0.60)	0.000 (1.0)	-0.061 (0.60)	0.000 (1.0)	0.000 (1.0)	$+0.061$ (0.60)
PATHOGAIT‡	$+0.021$	-0.002	-0.070	-0.052	-0.024	$+0.382$
ASD-Kinect	-0.023 (0.70)	-0.125 (0.25)	-0.125 (0.24)	-0.030 (0.89)	-0.030 (0.89)	0.000 (1.0)

CARE-PD provides the clearest evidence that the summary contains clip-specific signal: the matched summary condition exceeds both the no-evidence and shuffled-evidence controls. It nevertheless remains below the 0.659 logistic-reference macro-F1. GMFCS improves under SFT but remains sensitive to candidate order. On PARKINSONISM, z-features are substantially better than summaries within the tested model, while the five-subject uncertainty remains wide. ASD-Kinect yields no detectable benefit from summary evidence or SFT in this small test.

S1.9 In-Distribution Adaptation and Subject-Level Uncertainty

Table S11: Supervision ladder on the larger-test records. Macro-F1 / κ. Few-shot uses Qwen2.5-72B; zero-shot and LoRA-SFT use Qwen2.5-7B. ATAXIA/GMFCS/PARKINSONISM/PATHOGAIT test clips are 1,647/866/120/2,181; PATHOGAIT few-shot uses 600. GMFCS SFT and references use corrected participants; its matching 7B zero-shot anchor is unavailable.

Condition	ATAXIA	GMFCS	PARKINSONISM	PATHOGAIT
Zero-shot summary, 7B	0.452 / 0.046	–	0.400 / -0.045	0.087 / -0.024
Best few-shot summary	0.697 / 0.395	–	0.561 / 0.188	0.185 / 0.081
LoRA-SFT summary, 7B	0.751 / 0.507	0.259 / 0.043	0.454 / 0.065	0.609 / 0.511
LoRA-SFT feature view, 7B	0.762 / 0.528	–	–	0.687 / 0.611
Logistic reference	0.763 / 0.533	0.395 / 0.241	0.613 / 0.226	0.662 / 0.589
Linear-SVM reference	0.759 / 0.524	0.412 / 0.271	0.583 / 0.168	0.644 / 0.585

Supervision recovers substantial in-distribution signal on ATAXIA and PATHOGAIT. On ATAXIA, feature-view SFT and logistic regression differ by -0.001 macro-F1 under a paired subject-cluster bootstrap, with 95% CI $[-0.044, 0.033]$ and $p = 0.97$ across the three test participants. This supports comparability on the tested split, not clinical equivalence. On the corrected 866-clip GMFCS test, the archived paired summary-SFT minus LR difference is -0.137 macro-F1, with 95% participant-cluster interval $[-0.186, -0.085]$ and paired bootstrap tail area 0.001 over 166 participants. This archived comparison is not independently regenerated without the reference-model per-record predictions and source features. PARKINSONISM's SFT interval is wide (0.298–0.659) because only five participants are tested.

PATHOGAIT cannot support a subject-level difference interval. The two participant-specific macro-F1 values for feature-view SFT versus logistic regression are 0.720 versus 0.608 for one participant and 0.661 versus 0.673 for the other. The direction therefore reverses across participants. The aggregate feature-SFT score of 0.687 is described as comparable in point estimate to the linear references, not as evidence of population-level superiority.

S1.9.1 Participant-Level Reference Checks for the Shared Conditions

Table S12: Shared-condition leave-one-subject-out reference checks. leave-one-subject-out (LOSO) trains the logistic reference on all but one subject and evaluates the held-out subject, repeating over the available cohort. These checks concern recoverable feature signal, not LLM performance.

Resource	Subjects	LOSO macro-F1	95% subject CI
ATAXIA	20	0.950	[0.874, 1.000]
PARKINSONISM	23	0.561	[0.448, 0.659]
PATHOGAIT	10	0.743	[0.672, 0.810]
ASD-Kinect	100	0.780	[0.690, 0.859]

The LOSO checks show that the deterministic feature representation carries subject-level signal beyond one frozen split on ATAXIA, PATHOGAIT, and ASD-Kinect. PARKINSONISM remains a boundary condition: its point estimate is modest and the interval is broad.

S1.9.2 Participant-Level Reference Checks for the Model/Adaptation Records

Table S13 reports LR on the fixed model/adaptation test sets and on the corresponding pooled participant cohorts using leave-one-subject-out (LOSO). Test-set and pooled-cohort results have different evaluated participants and are not paired estimates. Their feature layouts and participant inventories also differ from the shared-condition LOSO check above. The corrected GMFCS fixed-test score and interval use the repaired split; the older permutation and pooled-cohort LOSO records are preserved only as historical diagnostics.

Table S13: Fixed-test and pooled-cohort LR checks for the model/adaptation experiments. Macro-F1 with participant-level intervals. Permutation values refer to the fixed-test analysis; "–" is unavailable.

Resource	Frozen test	Permutation p	LOSO	Subjects: frozen / LOSO
GMFCS	0.395 [0.350, 0.438]	–[‡]	0.406 [0.374, 0.436][‡]	166 / 344
ATAXIA	0.763 [0.625, 0.902]	0.0002	0.823 [0.757, 0.879]	3 / 20
PARKINSONISM	0.613 [0.402, 0.713]	0.0120	0.586 [0.525, 0.627]	5 / 26
PATHOGAIT	0.662 [0.662, 0.662][†]	–	0.609 [0.534, 0.674]	2 / 10
ASD-Kinect	–	–	–	–
CARE-PD	–	–	–	–

[†]PATHOGAIT has only two frozen-test participants. Its participant-cluster interval is degenerate and is reported for transparency rather than interpreted as population-level uncertainty. No frozen-test permutation claim is made for this condition. [‡]A corrected GMFCS permutation result is not recovered. Its LOSO value concerns the pooled 344-participant cohort and does not use the repaired frozen split.

S1.10 Tested Cross-Condition Transfer

Transfer is evaluated only between ATAXIA and PATHOGAIT because both can be harmonized to the binary task normal versus abnormal gait. PATHOGAIT's normal class maps to normal and its five performed pathological patterns map to abnormal; the target test split then contains 240 normal and 1,941 abnormal clips. GMFCS is ordinal without a healthy class, and PARKINSONISM predicts severity rather than normality, so neither enters this experiment.

Source-dataset training statistics standardize the named features. No target labels, target moments, or target examples are used. ATAXIA uses MediaPipe pose conventions and PATHOGAIT uses Kinect conventions; thus the experiment jointly changes gait condition, sensing pipeline, skeleton layout, and cohort. It is a deliberately difficult dataset-pair transfer test, not an isolation of disease shift alone.

Table S14: Bidirectional ATAXIA–PATHOGAIT transfer. Accuracy / macro-F1 / Cohen's κ after harmonizing the task to normal versus abnormal. The target-majority baseline is computed in the target label distribution.

Training resource	Evaluation resource	Model result	Target-majority baseline
ATAXIA	ATAXIA (in distribution)	– / – / 0.549	–
ATAXIA	PATHOGAIT (transfer)	0.656 / 0.496 / 0.057	0.890 / 0.471 / 0.000
PATHOGAIT	PATHOGAIT (in distribution)	0.984 / 0.959 / 0.917	–
PATHOGAIT	ATAXIA (transfer)	0.686 / 0.407 / 0.000	0.686 / 0.407 / 0.000

ATAXIA-to-PATHOGAIT is 23.4 percentage points below the target majority baseline in accuracy and only 0.025 above it in macro-F1, with $\kappa = 0.057$. PATHOGAIT-to-ATAXIA exactly reproduces the target majority result. Reliable transfer is therefore not observed on this tested bidirectional dataset pair. The result does not establish a general inability to transfer across clinical gait conditions.

S1.11 Per-Class Performance and Confusion Matrices

All tables and figures in this section are recomputed from the released shared-condition prediction files for Qwen2.5-7B-Instruct. They compare zero-shot motion summaries, zero-shot z-scored feature descriptions, and motion-summary LoRA-SFT. Rows of every confusion matrix are true classes; columns are predicted classes. Each cell shows the integer count and the percentage within its true class. The complete candidate set is used when computing macro-F1, even if a class is never predicted.

Table S15: Binary classification metrics. Sensitivity is recall of the explicitly named positive class, and specificity is recall of its complement. Case-sensitive source label identities are preserved.

Dataset	Condition	Positive class	Sensitivity	Specificity	Balanced acc.
ASD-Kinect	Zero-shot summary	ASD	0.000	0.900	0.450
ASD-Kinect	Zero-shot z-features	ASD	0.100	1.000	0.550
ASD-Kinect	LoRA-SFT summary	ASD	0.900	0.000	0.450
ATAXIA	Zero-shot summary	mimicked_ataxia	0.000	0.750	0.375
ATAXIA	Zero-shot z-features	mimicked_ataxia	0.000	1.000	0.500
ATAXIA	LoRA-SFT summary	mimicked_ataxia	1.000	0.000	0.500
PARKINSONISM	Zero-shot summary	severe	0.067	1.000	0.533
PARKINSONISM	Zero-shot z-features	severe	0.667	0.857	0.762
PARKINSONISM	LoRA-SFT summary	severe	0.867	0.286	0.576

The binary tables expose clinically relevant asymmetries hidden by accuracy. PARKINSONISM zero-shot summary sensitivity to severe clips is only 0.067 despite 0.517 accuracy; the z-feature view increases severe sensitivity to 0.667 while retaining 0.857 specificity. ATAXIA and ASD-Kinect each show a condition-dependent single-class collapse on their small shared-condition splits, making a single accuracy value unsuitable for interpretation.

Table S16: Ordinal error structure. MAE and quadratic-weighted κ use the ordered numeric labels.

Dataset	Condition	Macro-F1	MAE	Quadratic κ
CARE-PD	Zero-shot summary	0.321	0.717	0.217
CARE-PD	Zero-shot z-features	0.133	1.211	0.004
CARE-PD	LoRA-SFT summary	0.445	0.651	0.331
GMFCS	Zero-shot summary	0.265	0.906	0.129
GMFCS	Zero-shot z-features	0.150	1.141	0.057
GMFCS	LoRA-SFT summary	0.467	0.420	0.618

On GMFCS, SFT reduces MAE from 0.906 to 0.420 and raises quadratic-weighted κ from 0.129 to 0.618. Neither the zero-shot summary nor the LoRA-SFT summary condition correctly predicts a level-4 clip; the z-scored feature condition correctly predicts one of the three level-4 clips. This support remains too small for stable class-specific inference.

Table S17: ATAXIA per-class results. Precision, recall, and F1 are recomputed from committed predictions; all task classes are retained.

Condition	Class	Support	Precision	Recall	F1
Zero-shot summary	mimicked_ataxia	4	0.000	0.000	0.000
Zero-shot summary	normal	4	0.429	0.750	0.545
Zero-shot z-features	mimicked_ataxia	4	0.000	0.000	0.000
Zero-shot z-features	normal	4	0.500	1.000	0.667
LoRA-SFT summary	mimicked_ataxia	4	0.500	1.000	0.667
LoRA-SFT summary	normal	4	0.000	0.000	0.000

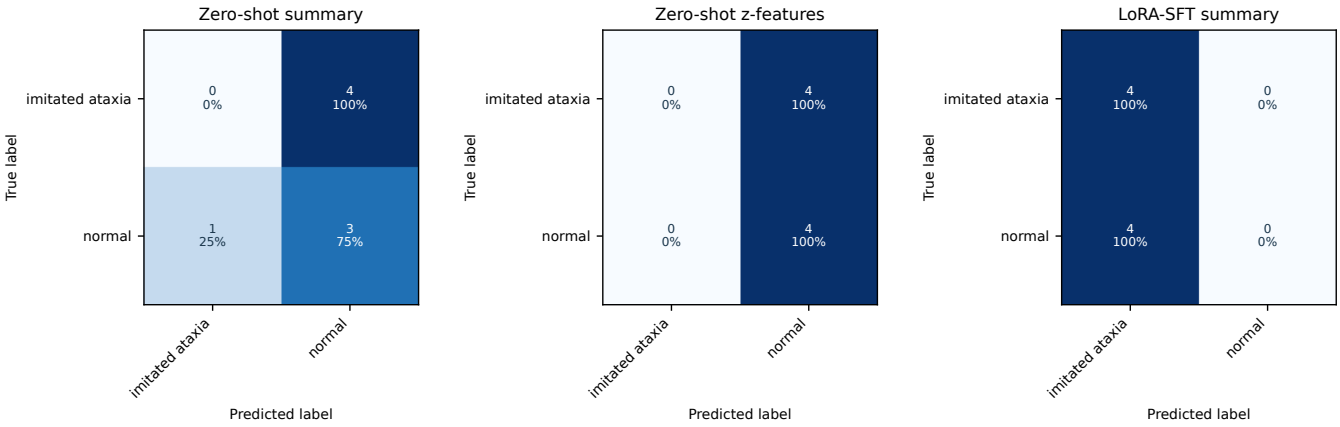


Figure S1: ATAXIA confusion matrices. Eight clips from four participants make these descriptive rather than rank-defining.

Table S18: GMFCS per-class results. Precision, recall, and F1 are recomputed from committed predictions; all task classes are retained.

Condition	Class	Support	Precision	Recall	F1
Zero-shot summary	1	98	0.420	0.347	0.380
Zero-shot summary	2	97	0.361	0.134	0.195
Zero-shot summary	3	57	0.352	0.772	0.484
Zero-shot summary	4	3	0.000	0.000	0.000
Zero-shot z-features	1	98	0.000	0.000	0.000
Zero-shot z-features	2	97	0.293	0.124	0.174
Zero-shot z-features	3	57	0.237	0.772	0.362
Zero-shot z-features	4	3	0.036	0.333	0.065
LoRA-SFT summary	1	98	0.611	0.592	0.601
LoRA-SFT summary	2	97	0.514	0.557	0.535
LoRA-SFT summary	3	57	0.745	0.719	0.732
LoRA-SFT summary	4	3	0.000	0.000	0.000

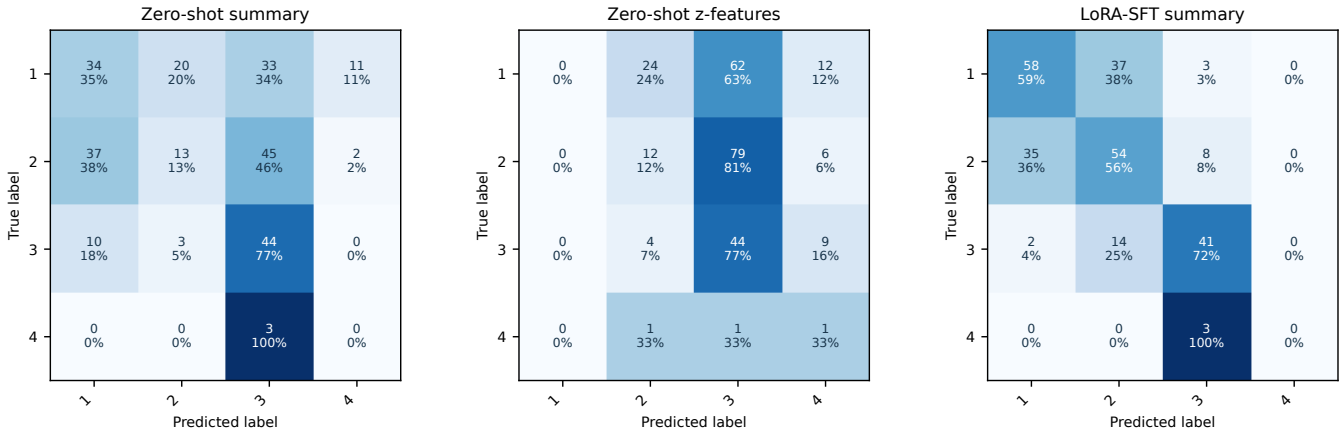


Figure S2: GMFCS confusion matrices. SFT improves levels 1–3 but the three level-4 clips are all predicted as level 3.

Table S19: PARKINSONISM per-class results. Precision, recall, and F1 are recomputed from committed predictions; all task classes are retained.

Condition	Class	Support	Precision	Recall	F1
Zero-shot summary	mild	14	0.500	1.000	0.667
Zero-shot summary	severe	15	1.000	0.067	0.125
Zero-shot z-features	mild	14	0.706	0.857	0.774
Zero-shot z-features	severe	15	0.833	0.667	0.741
LoRA-SFT summary	mild	14	0.667	0.286	0.400
LoRA-SFT summary	severe	15	0.565	0.867	0.684

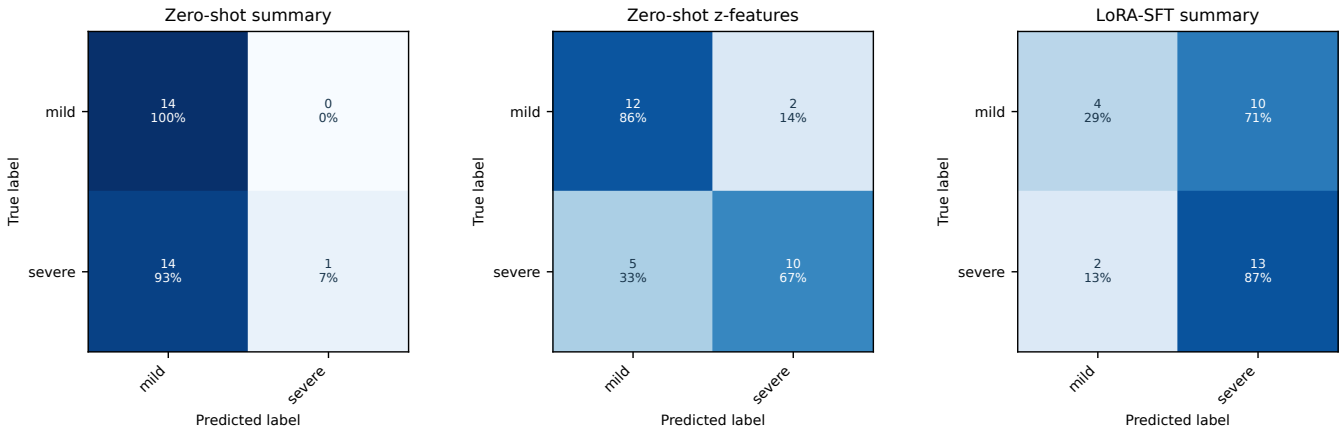


Figure S3: PARKINSONISM confusion matrices. The z-feature view provides the most balanced class recovery in this five-subject test.

Table S20: PATHOGAIT per-class results. Precision, recall, and F1 are recomputed from committed predictions; all task classes are retained.

Condition	Class	Support	Precision	Recall	F1
Zero-shot summary	antalgic	240	0.000	0.000	0.000
Zero-shot summary	lurch	240	0.027	0.004	0.007
Zero-shot summary	normal	240	0.184	1.000	0.310
Zero-shot summary	steppage	240	0.000	0.000	0.000
Zero-shot summary	stiff_legged	240	0.000	0.000	0.000
Zero-shot summary	trendelenburg	240	0.174	0.067	0.096
Zero-shot z-features	antalgic	240	0.038	0.004	0.008
Zero-shot z-features	lurch	240	0.114	0.017	0.029
Zero-shot z-features	normal	240	0.173	0.700	0.278
Zero-shot z-features	steppage	240	0.117	0.087	0.100
Zero-shot z-features	stiff_legged	240	0.000	0.000	0.000
Zero-shot z-features	trendelenburg	240	0.428	0.408	0.418
LoRA-SFT summary	antalgic	240	0.462	0.350	0.398
LoRA-SFT summary	lurch	240	0.379	0.371	0.375
LoRA-SFT summary	normal	240	0.720	0.558	0.629
LoRA-SFT summary	steppage	240	0.319	0.438	0.369
LoRA-SFT summary	stiff_legged	240	0.640	0.808	0.715
LoRA-SFT summary	trendelenburg	240	0.239	0.204	0.220

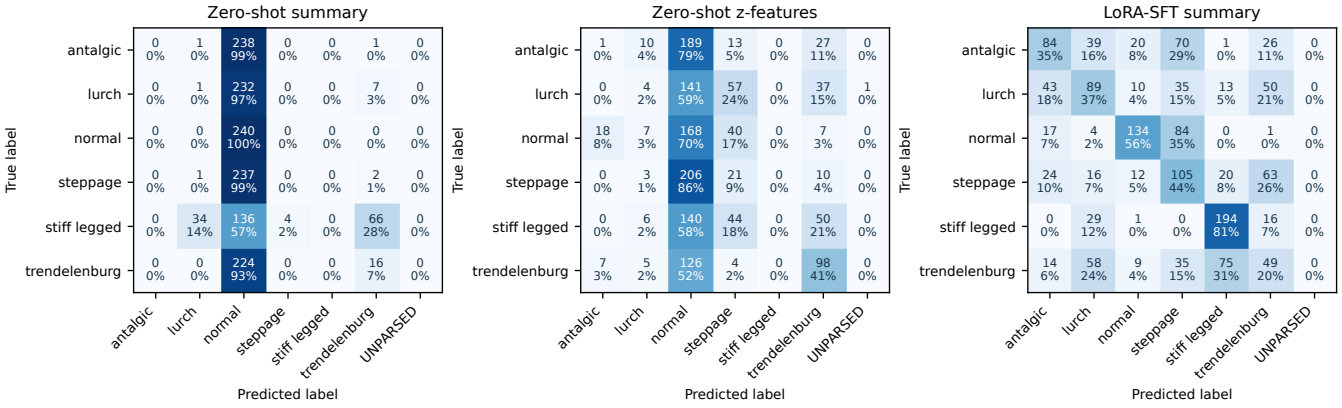


Figure S4: PATHOGAIT confusion matrices. Zero-shot summary predictions concentrate on normal; SFT distributes predictions across all six patterns but remains uneven, particularly for Trendelenburg gait. The z-feature condition contains one unparsed lurch output, shown as an error.

Table S21: ASD-Kinect per-class results. Precision, recall, and F1 are recomputed from committed predictions; all task classes are retained.

Condition	Class	Support	Precision	Recall	F1
Zero-shot summary	ASD	10	0.000	0.000	0.000
Zero-shot summary	typically_developing	10	0.474	0.900	0.621
Zero-shot z-features	ASD	10	1.000	0.100	0.182
Zero-shot z-features	typically_developing	10	0.526	1.000	0.690
LoRA-SFT summary	ASD	10	0.474	0.900	0.621
LoRA-SFT summary	typically_developing	10	0.000	0.000	0.000

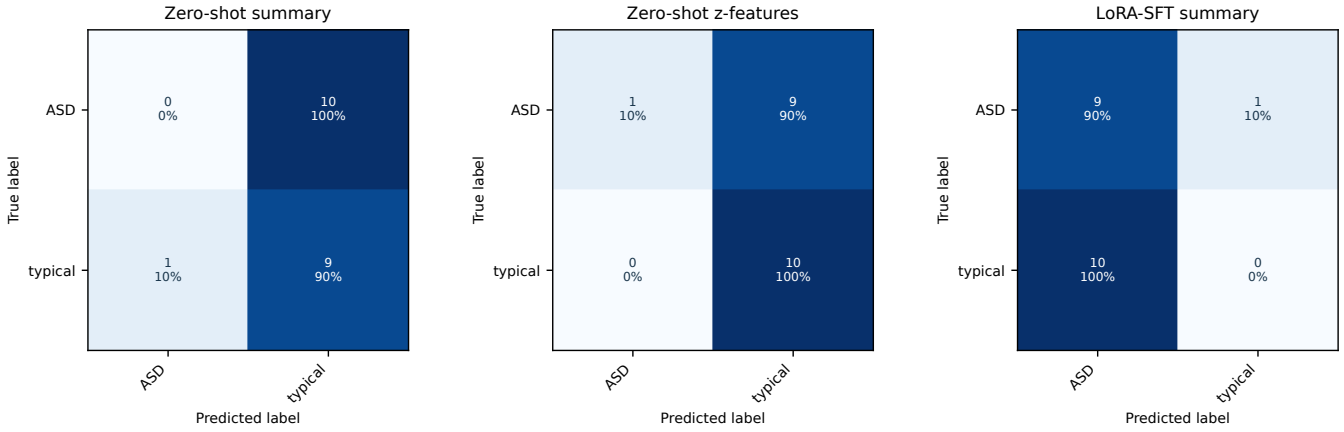


Figure S5: ASD-Kinect confusion matrices. One clip per participant yields 20 independent observations, but the small class supports make all estimates coarse.

Table S22: CARE-PD per-class results. Precision, recall, and F1 are recomputed from committed predictions; all task classes are retained.

Condition	Class	Support	Precision	Recall	F1
Zero-shot summary	0	240	0.528	0.471	0.498
Zero-shot summary	1	247	0.455	0.490	0.472
Zero-shot summary	2	160	0.292	0.338	0.313
Zero-shot summary	3	18	0.000	0.000	0.000
Zero-shot z-features	0	240	0.000	0.000	0.000
Zero-shot z-features	1	247	0.310	0.073	0.118
Zero-shot z-features	2	160	0.275	0.856	0.416
Zero-shot z-features	3	18	0.000	0.000	0.000
LoRA-SFT summary	0	240	0.519	0.683	0.590
LoRA-SFT summary	1	247	0.414	0.441	0.427
LoRA-SFT summary	2	160	0.304	0.150	0.201
LoRA-SFT summary	3	18	1.000	0.389	0.560

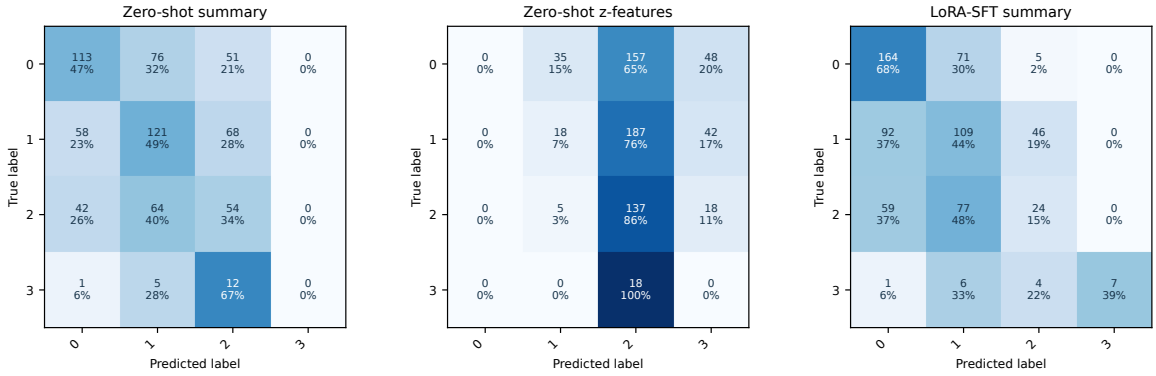


Figure S6: CARE-PD UPDRS-gait confusion matrices. The evaluation is restricted to scores 0–3 from one task. Score 3 has 18 clips and should not be interpreted as a precise severity-specific estimate.

The six confusion analyses refine the aggregate conclusion. Zero-shot failure is frequently a class-allocation failure rather than uniformly low recall: ATAXIA, ASD-Kinect, and PARKINSONISM each approach a single-class decision under at least one view. GMFCS and CARE-PD errors are predominantly adjacent or near-adjacent under SFT, which is reflected by improved ordinal metrics even when macro-F1 remains below the feature references. PATHOGAIT SFT recovers substantial normal and stiff-legged performance but not a balanced six-class mapping.

S1.12 Reproducibility Contract and Evidence Limits

The release data model records participant manifests, native/canonical poses, raw/train-scaled features, label-free summaries, prompts, and per-clip predictions. The 55-file shared-condition collection contains raw outputs, summaries, scoring, control, and training code and reference/statistical records. The separate split-repair evidence contains GMFCS participant maps and retrained-model predictions and is not part of that count. Complete source poses, feature/summary input layers, split-generation inputs, adapters, and training logs remain external. The supplied audit checks available predictions and test-record alignment without claiming that full retraining has been reproduced.

The following boundaries apply to all reported conclusions:

1. GaitEval measures dataset-specified classification grounded in pose-derived evidence; it does not validate autonomous diagnosis, treatment choice, or clinical deployment.
2. ATAXIA and PATHOGAIT contain performed rather than patient gait; CARE-PD uses a task-specific clinical subset, and ASD-Kinect has the source-provenance caveat stated above. These are population limitations within one six-dataset benchmark.
3. Skeleton estimation removes appearance cues but introduces sensor-, estimator-, and layout-specific noise. The deterministic feature set is interpretable but incomplete, and unsigned asymmetry loses laterality.
4. Test cohorts are small for ATAXIA, PARKINSONISM, PATHOGAIT, ASD-Kinect, and the highest ordinal severity classes. Subject-level intervals, per-class support, and confusion matrices must accompany point estimates.
5. In-distribution SFT demonstrates learning of supplied task mappings. It does not by itself show acquisition of transferable clinical knowledge.
6. Cross-condition evidence is restricted to the harmonized ATAXIA–PATHOGAIT pair and combines condition, cohort, sensor, and representation shift. No broader cross-disease transfer conclusion follows.

S1.13 Six-Dataset Experimental Coverage

All six datasets are benchmark members under one evaluation protocol. Seven shared Qwen-7B conditions give 42 prediction-backed cells. Two additional raw-feature cells and ten ASD-Kinect/CARE-PD model-comparison cells, plus a separate 12-class CARE-PD diagnostic, give 55 files in this collection. Other model/view/adaptation results are recorded on the test populations stated in their captions. Coordinate prompting, few-shot, feature-SFT, and transfer have the coverage in Table S23; benchmark membership does not imply a complete Cartesian product of configurations.

Table S23: Experimental coverage across six datasets. "Yes" means a result exists on the stated test set; "–" means no result; "Unverified" means the corrected records needed to validate a historical result are missing. All six datasets have the seven Qwen-7B conditions and matched LR/SVM. Test counts for each experiment are stated in the result captions and supplemental evaluation-record inventory. The five-model column excludes the Qwen-7B anchor.

Dataset	Five families	ZS summary/z	ZS raw feat.	Summary SFT	LR/SVM+controls	Few-shot	Feature SFT	Coordinates	Transfer
ASD-Kinect	Yes	Yes	Yes	Yes	Yes	–	–	–	–
ATAXIA	Yes	Yes	–	Yes	Yes	Yes	Yes	Yes	Tested pair
CARE-PD	Yes	Yes	Yes	Yes	Yes	–	–	–	–
GMFCS	Yes	Yes	–	Yes	Yes	–	–	Unverified	–
PARKINSONISM	Yes	Yes	–	Yes	Yes	Yes	–	Normalized	–
PATHOGAIT	Yes	Yes	–	Yes	Yes	Yes	Yes	Yes	Tested pair

Table S24: Same-feature references for the seven shared conditions. All six datasets use the features and evaluation records specified for the shared-condition comparisons. CARE-PD is the four-class UPDRS gait task with 665 test clips. The LR/SVM inputs match the corresponding named-feature LLM inputs; n and Subjects identify the evaluated records.

Dataset	n	Subjects	Chance mF1	LR Acc.	LR mF1	LR κ	SVM Acc.	SVM mF1	SVM κ
ASD-Kinect	20	20	0.490	0.700	0.697	0.400	0.750	0.744	0.500
ATAXIA	8	4	0.488	0.875	0.873	0.750	0.875	0.873	0.750
CARE-PD	665	24	0.250	0.621	0.659	0.437	0.621	0.650	0.435
GMFCS	255	149	0.250	0.639	0.579	0.458	0.647	0.571	0.469
PARKINSONISM	29	5	0.494	0.586	0.561	0.187	0.586	0.574	0.183
PATHOGAIT	1440	2	0.166	0.881	0.881	0.858	0.897	0.897	0.877

Table S25: Six-dataset input-view comparison with Qwen2.5-7B. Accuracy / macro-F1 / κ on identical within-dataset test records: 20/8/665/255/29/1,440 clips in dataset order. Raw-feature runs exist only where shown; "–" is an unexecuted configuration. Coordinate prompting was not run on these test sets.

Dataset	Motion summary	Raw named features	Z-scored named features
ASD-Kinect	0.450 / 0.310 / −0.100	0.500 / 0.333 / 0.000	0.550 / 0.436 / 0.100
ATAXIA	0.375 / 0.273 / −0.250	–	0.500 / 0.333 / 0.000
CARE-PD	0.433 / 0.321 / 0.152	0.332 / 0.221 / 0.039	0.233 / 0.133 / 0.020
GMFCS	0.357 / 0.265 / 0.099	–	0.224 / 0.150 / −0.003
PARKINSONISM	0.517 / 0.396 / 0.065	–	0.759 / 0.757 / 0.520
PATHOGAIT	0.178 / 0.069 / 0.014	–	0.203 / 0.139 / 0.043

Known accepted-version discrepancies. The GMFCS source described 899 original test clips, 885 in a resource summary, and 866 after correction. Released maps resolve the original counts as 1,574/473/899 and the repaired counts as 1,602/478/866. Verified corrected values are used for model comparison, SFT, and references. Newly recovered 7B view numbers are shown in red for author review, with their distinct provenance disclosed; matching per-record outputs remain unavailable and no provisional cell is used in scientific comparisons. The seven-condition experiment has 255 test clips from 149 participants and an incomplete released training manifest. CARE-PD primary references use 665 clips/four labels; the 970-clip/12-label reference is a different task and excluded. Normalized result records and generated checks take precedence over stale source Markdown.

Prompt and parser settings. The recorded `view_run.py` template contains a 3D-skeleton/biomechanical-dimension preamble; `v21_llm_runner.py` renders the seven-condition evidence with a shorter skeleton-sequence preamble. The former parser has a unique truncated-label-prefix fallback; the latter requires complete boundary-anchored candidates. These implementation settings remain attached to their actual outputs. Scores are not substituted across different prompts, test records, or parsers, and missing experiments remain unreported.

Evidence assets still unavailable. The collection includes committed results and corrective reruns. This revision executes no new training or inference. Complete records, pose/feature/summary layers, and training artifacts are required for end-to-end reproduction. A canonical 19-joint data layer does not establish that the 12-joint coordinate-input configuration was run on every evaluated record set.

Recomputed binary-label identity. ASD-Kinect uses the positive label `ASD`. Recomputing from the committed shared-condition predictions gives summary sensitivity/specificity 0.000/0.900, z-feature 0.100/1.000, and summary-SFT 0.900/0.000. These numerical values agree with the accepted binary-metric table; the regenerated table preserves them and standardizes the displayed label identity. This is a consistency check, not a correction to the accepted binary scores or an extra experiment.

S2 Complete Results by Experimental Setting

Table S26: Five-model zero-shot motion-summary benchmark. Accuracy / macro-F1 / Cohen's κ for ASD-Kinect/ATAXIA/CARE-PD/GMFCS/PARKINSONISM/PATHOGAIT, respectively. These test sets also define the supervision ladder; their sizes differ from the seven-condition comparison where indicated. Bold marks the highest macro-F1 per dataset; invalid outputs count as errors.

Dataset (Maj. Acc.)	Llama-3.3-70B	Qwen2.5-72B	Gemma-3-27B	GLM-4-32B	DeepSeek-R1-32B
ASD-Kinect (0.500)	0.500 / 0.451 / 0.000	0.450 / 0.373 / −0.100	0.450 / 0.310 / −0.100	0.450 / 0.373 / −0.100	0.600 / **0.583** / 0.200
ATAXIA (0.686)	0.686 / 0.407 / 0.000	0.685 / 0.406 / −0.002	0.686 / 0.407 / 0.000	0.686 / 0.407 / 0.000	0.684 / **0.408** / −0.002
CARE-PD (0.361)	0.441 / 0.262 / 0.119	0.445 / 0.268 / 0.129	0.439 / 0.249 / 0.113	0.441 / **0.350** / 0.124	0.398 / 0.295 / 0.099
GMFCS (0.356)	0.353 / 0.212 / 0.025	0.372 / **0.222** / 0.043	0.313 / 0.163 / 0.030	0.269 / 0.127 / −0.000	0.311 / 0.191 / 0.014
PARKINSONISM (0.583)	0.583 / 0.368 / 0.000	0.575 / **0.472** / 0.035	0.583 / 0.368 / 0.000	0.583 / 0.368 / 0.000	0.583 / 0.368 / 0.000
PATHOGAIT (0.226)	0.159 / 0.062 / −0.010	0.155 / 0.055 / −0.009	0.165 / 0.047 / −0.001	0.141 / 0.069 / −0.069	0.166 / **0.094** / −0.007

Table S27: Shared six-dataset evidence-use and adaptation evaluation. Macro-F1 point estimates use the fixed task label set. Bracketed participant-cluster intervals and paired bootstrap tail areas p are archived statistics with the historical resampling convention described in Methods; they are not newly recomputed fixed-label intervals. Δ_{ShufEv} is summary zero-shot minus shuffled evidence; Δ_{SFT} is summary-SFT minus summary zero-shot within Qwen2.5-7B. PATHOGAIT has only two test participants, so its values are descriptive and intervals/p are withheld.

Dataset	Task	Clips (subjects)	Chance	LR [95% CI]	7B ZS [95% CI]	Δ_{ShufEv} (p)	7B SFT (Δ_{SFT}, p)
ASD-Kinect	binary	20 (20)	.490	.697 [.479, .890]	.310 [.200, .394]	−.125 (.25)	.310 (.000, 1.0)
ATAXIA	binary, imitated	8 (4)	.488	.873 [.564, 1.000]	.273 [.111, .333]	.000 (1.0)	.333 (+.061, .60)
CARE-PD	UPDRS gait 0–3	665 (24)	.250	.659 [.407, .742]	.321 [.262, .404]	+.065 (.029)	.445 (+.124, .43)
GMFCS	four-level ordinal	255 (149)	.250	.579 [.474, .662]	.265 [.217, .308]	+.052 (.089)	.467 (+.202, .001)
PARKINSONISM	binary severity	29 (5)	.494	.561 [.416, .654]	.396 [.222, .530]	+.086 (.14)	.542 (+.146, .47)
PATHOGAIT	six-class, simulated	1440 (2)	.166	.881	.069	−.002	.451 (+.382)

Table S28: Four input-view families and a feature-scaling control. Accuracy / macro-F1 / κ. Z-scoring is an encoding of the named features. "–" means no recoverable numeric result.

Input view	ASD-Kinect	ATAXIA	CARE-PD	GMFCS	PARKINSONISM	PATHOGAIT
Raw coordinates[†]	–	0.573 / 0.323 / 0.009	–	–	–	0.093 / 0.083 / 0.038
Normalized coordinates	–	0.316 / 0.244 / 0.002	–	0.282 / 0.110 / 0.000	0.442 / 0.441 / −0.095	0.110 / 0.034 / 0.000
Motion summary	0.450 / 0.310 / −0.100	0.692 / 0.452 / 0.046	0.433 / 0.321 / 0.152	0.259 / 0.113 / −0.009	0.550 / 0.400 / −0.045	0.142 / 0.087 / −0.024
Named feature vector	0.500 / 0.333 / 0.000	0.382 / 0.347 / 0.057	0.332 / 0.221 / 0.039	0.343 / 0.245 / 0.007	0.417 / 0.294 / 0.000	0.110 / 0.033 / 0.000
Z-scored named feature vector	0.550 / 0.436 / 0.100	0.583 / **0.495** / −0.008	0.233 / 0.133 / 0.020	0.185 / 0.100 / −0.049	0.508 / **0.493** / −0.014	0.150 / **0.130** / −0.011

Table S29: Six-dataset input-view comparison with Qwen2.5-7B. Accuracy / macro-F1 / κ on identical within-dataset test records: 20/8/665/255/29/1,440 clips in dataset order. Raw-feature runs exist only where shown; "–" is an unexecuted configuration. Coordinate prompting was not run on these test sets.

Dataset	Motion summary	Raw named features	Z-scored named features
ASD-Kinect	0.450 / 0.310 / −0.100	0.500 / 0.333 / 0.000	0.550 / 0.436 / 0.100
ATAXIA	0.375 / 0.273 / −0.250	–	0.500 / 0.333 / 0.000
CARE-PD	0.433 / 0.321 / 0.152	0.332 / 0.221 / 0.039	0.233 / 0.133 / 0.020
GMFCS	0.357 / 0.265 / 0.099	–	0.224 / 0.150 / −0.003
PARKINSONISM	0.517 / 0.396 / 0.065	–	0.759 / 0.757 / 0.520
PATHOGAIT	0.178 / 0.069 / 0.014	–	0.203 / 0.139 / 0.043

Table S30: Same-feature linear references for the model/view experiments. Accuracy, macro-F1, and Cohen's κ; test sizes are given in the final column. Chance mF1 is the recorded stratified-chance reference. Bold marks the stronger linear reference per metric. These references match the feature inputs and test populations of the supervision ladder; the next table gives references for the seven shared conditions.

Dataset	Classes	Maj. Acc.	Chance mF1	LR Acc.	LR mF1	LR κ	SVM Acc.	SVM mF1	SVM κ	n_{te}
ASD-Kinect	2	0.500	0.490	0.700	0.697	0.400	0.750	0.744	0.500	20
ATAXIA	2	0.686	0.498	**0.778**	**0.763**	**0.533**	0.774	0.759	0.524	1647
CARE-PD	4	0.361	0.250	0.621	0.659	0.437	0.621	0.650	0.435	665
GMFCS	4	0.356	0.250	0.461	0.395	0.241	**0.492**	**0.412**	**0.271**	866
PARKINSONISM	2	0.583	0.499	**0.625**	**0.613**	**0.226**	0.600	0.583	0.168	120
PATHOGAIT	6	0.226	0.166	**0.659**	**0.662**	**0.589**	0.656	0.644	0.585	2181

Table S31: Same-feature references for the seven shared conditions. All six datasets use the features and evaluation records specified for the shared-condition comparisons. CARE-PD is the four-class UPDRS gait task with 665 test clips. The LR/SVM inputs match the corresponding named-feature LLM inputs; n and Subjects identify the evaluated records.

Dataset	n	Subjects	Chance mF1	LR Acc.	LR mF1	LR κ	SVM Acc.	SVM mF1	SVM κ
ASD-Kinect	20	20	0.490	0.700	0.697	0.400	0.750	0.744	0.500
ATAXIA	8	4	0.488	0.875	0.873	0.750	0.875	0.873	0.750
CARE-PD	665	24	0.250	0.621	0.659	0.437	0.621	0.650	0.435
GMFCS	255	149	0.250	0.639	0.579	0.458	0.647	0.571	0.469
PARKINSONISM	29	5	0.494	0.586	0.561	0.187	0.586	0.574	0.183
PATHOGAIT	1440	2	0.166	0.881	0.881	0.858	0.897	0.897	0.877

Table S32: Task-specific adaptation ladder. LLM cells are macro-F1 / Cohen's κ; references are macro-F1. Zero-shot/few-shot use Qwen2.5-72B, while the 7B anchor and SFT use Qwen2.5-7B. Test clips in dataset order are 20/1,647/665/866/120/2,181. PATHOGAIT few-shot instead uses 600 clips, so that change is not a paired full-test gain. GMFCS uses corrected training/test partitions; its matching 7B zero-shot anchor is unverified and withheld. "–" denotes unavailable configurations. Bold feature-SFT matches or exceeds the stronger reference point estimate.

Dataset	Zero-shot	Zero-shot (7B)	Few-shot	SFT: summary	SFT: features	SVM	LogReg
ASD-Kinect	0.373 / −0.100	0.310 / −0.100	–	0.310 / −0.100	–	0.744	0.697
ATAXIA	0.406 / −0.002	0.452 / 0.046	0.697 / 0.395	0.751 / 0.507	0.762 / 0.528	0.759	0.763
CARE-PD	0.268 / 0.129	0.321 / 0.152	–	0.445 / 0.168	–	0.650	0.659
GMFCS	0.222 / 0.043	–	–	0.259 / 0.043	–	0.412	0.395
PARKINSONISM	0.472 / 0.035	0.400 / −0.045	0.561 / 0.188	0.454 / 0.065	–	0.583	0.613
PATHOGAIT	0.055 / −0.009	0.087 / −0.024	0.185 / 0.081	0.609 / 0.511	**0.687 / 0.611**	0.644	0.662

Table S33: Bidirectional ATAXIA–PATHOGAIT transfer. Models use z-scored feature descriptions; target-majority values are accuracy / macro-F1. Target tests have 2,181 PATHOGAIT clips or 1,647 ATAXIA clips. PATHOGAIT is binarized, so its in-distribution κ is not comparable with the six-class adaptation result. OOD denotes target-dataset evaluation. Only this task-compatible pair was tested.

Train → Test	In-dist. κ	OOD Acc. / mF1 / κ	Target majority
ATAXIA → PATHOGAIT	0.549	0.656 / 0.496 / 0.057	0.890 / 0.471
PATHOGAIT → ATAXIA	0.917	0.686 / 0.407 / 0.000	0.686 / 0.407

S3 Source examples and release boundary

Figure 1 illustrates source resources rather than the evaluated test clips. No RGB images are supplied to the text-only models. The source publications and their original references in the main paper remain the attribution basis. Derived plots and vector diagrams are newly drawn; source-example crops retain their separate attribution and license notes in the project repository.

The six-dataset scoring package reproduces point estimates from committed predictions. Reported reference scores and archived cluster-bootstrap outputs are preserved as source measurements, not claimed to be retrained or independently regenerated without their inputs. Full source pose, feature, summary, and split files and adapters are required for end-to-end inference/training reproduction. The code validates participant separation when those split files are supplied and fails on missing, empty, or overlapping inputs.

Resource-example credits. ASD-Kinect uses a cropped frame at 0.5 s from the original `Svideo.avi`, child-ASD/1/video, in the official Zenodo archive. ATAXIA uses captured-video illustrations from Khalil et al., *Diagnosis of Cerebellar Ataxia Based on Gait Analysis Using Human Pose Estimation*, IEEE IECBES2022 (DOI:10.1109/IECBES54088.2022.10079396). CARE-PD uses a crop of the official benchmark input-mesh example. GMFCS uses a newly rendered skeleton from mobile-gaitlab's public demo keypoints (frame60), not confirmed as an evaluated participant. PARKINSONISM uses a cropped recording illustration from Deng et al., *npj Parkinson's Disease*10,122(2024), DOI:10.1038/s41531-024-00742-x (CC BY4.0). PATHOGAIT renders recorded joints from the official source repository. Crops and skeleton renderings were made for this overview. These are resource illustrations, not clip-matched model evidence. Source-paper image reuse terms and any unlicensed derived display must be confirmed before final distribution; dataset availability alone does not establish image permission.

S4 Reviewer-requested qualitative examples

Table S34: Prediction-backed qualitative decisions across all six datasets. Outputs are verbatim from committed logs. No matching pose or prompt is inferred.

Dataset / view	True label	Raw output	Correct
asd_kinect / zs	typically_developing	typically_developing	yes
asd_kinect / zs	ASD	typically_developing	no
ataxia / zs	normal	normal	yes
ataxia / zs	mimicked_ataxia	normal	no
care_pd / zs	0	UPDRS gait score 0	yes
care_pd / zs	1	UPDRS gait score 0	no
gmfcs / zs	3	GMFCS level 3	yes
gmfcs / zs	2	GMFCS level 1	no
parkinsonism / zs	severe	severe	yes
parkinsonism / zs	severe	mild	no
pathogait / zs	normal	normal	yes
pathogait / zs	antalgic	normal	no

For each dataset, the examples are deterministic selections from the committed predictions, showing one correct and one incorrect summary or z-feature decision where available. Raw model outputs are shown verbatim; true labels are evaluation annotations, not model-visible metadata. These diagnostics are not representative sampling and do not establish reasoning faithfulness. The original shared-condition input prompts and matching pose records are absent, so no numerical evidence is reconstructed or invented for these cases. Those outputs are therefore not complete matched-input panels.

S5 Recovered Corrective Experiments

This section incorporates corrective experiments that were already executed and released by the authors. It does not report new training performed for this revision. The correction package records source commits `2348b11` of the private working archive and `60f3503` of the public GaitEval repository. The verification script recomputes split intersections, exact test-ID alignment, raw-output parsing, SFT point metrics, and the corrected evaluation-record joins. Archived reference-model and bootstrap values are identified as such rather than represented as independently retrained here.

S5.1 GMFCS: Original and Corrected Participant Partitions

The original partition contains 1,574/473/899 train/validation/test clips. Its original-to-corrected index maps assign a participant to the first split in training–validation–test order in which that participant appears, and then move all that participant's clips to that split. The rule preserves source clips and yields 1,602/478/866 clips from 94/84/166 participants. Direct intersections of every pair of released participant sets are empty. In particular, the 866 corrected test records correspond to participants absent from both training and validation. The released SFT output IDs are exactly this test set and their true labels match the original indexed records.

The linear references and summary-SFT were rerun on corrected training data. Cached zero-shot summary generations were instead restricted to the corrected test IDs and re-parsed; this is appropriate for those deterministic, non-fitted summary prompts. The archived corrected SFT-minus-LR participant-bootstrap difference is -0.1367, interval $[-0.1862, -0.0853]$, paired tail area 0.0010. Source features and per-record reference predictions are needed to independently regenerate that bootstrap. This archived result is distinct from the independently rescored SFT point estimate.

Table S35: GMFCS original versus corrected split evidence. Values are accuracy / macro-F1 / κ. Reference rows and original rows are archived source measurements; the corrected SFT row is independently recomputed from the supplied generations. The original and corrected SFT rows have different test sets. These use 866 test clips; the seven-condition comparison uses 255.

Experiment	Original: 899 clips	Corrected: 866 clips
72B zero-shot summary	.3693 / .2182 / .0325	.3718 / .2216 / .0426
7B summary-SFT	.3626 / .2553 / .0298	.3730 / .2585 / .0431
Balanced LR	.4516 / .3875 / .2290	.4607 / .3952 / .2409
Balanced SVM	.4917 / .4101 / .2706	.4919 / .4121 / .2708

The corrected five-model rows consistently use 866 GMFCS test IDs. Recovered summary/raw-feature/z-feature aggregates refer to 899 clips; normalized-coordinate outputs also lack independently verified corrected IDs. The newly displayed red summary instead uses a logged 866-clip cross-stack run. All four red GMFCS view cells are provisional and excluded from matched-test comparisons. Corrected participant maps apply to the 866-clip experiment. The 255-clip/149-participant, 16-feature shared-condition experiment still requires its full training input manifest for independent separation checks. These two recorded test populations are never treated as interchangeable estimates.

7B view lineage audit. For this author-review draft, the source experiment log dated 2026-10-19 reports a separate 866-clip Qwen2.5-7B zero-shot summary run on the bigpurple serving stack: 0.2587 / 0.1129 / -0.0089. Its per-record outputs have not been recovered, so it is shown provisionally rather than substituted into the verified supervision ladder. The red raw/z-feature entries below use the historical 899-clip aggregates; the normalized-coordinate entry retains its unverified source-draft status. PATHOGAIT's red named-feature entry is recovered from `zeroshot-feat-qwen7b_pathological` in `r2_recompute_part1.json`: all 2,181 outputs are valid, all predict normal gait, and the aggregate is 0.110041 / 0.033044 / 0.000000. Its launcher specifies the base Qwen2.5-7B-Instruct model, test split, unscaled feature-vector input, flat prompt, and 16 output tokens. The previous table generator searched only the alternative `zs7b_feature_vector` name and omitted this run. The recovered `r2_recompute_part1.json` associates the following values with the original 899-clip split. Raw per-record generations for these view runs were not recovered, so a corrected 866-clip score cannot be reconstructed from aggregate statistics. In particular, the historical 0.110 summary macro-F1 must not serve as a same-test zero-shot anchor for corrected SFT at 0.2585. The normalized-coordinate cell (0.282 / 0.110 / 0.000) has only a source-draft corrected-split claim and is likewise unverified.

Table S36: Historical GMFCS 7B views, not corrected-split results. All three rows have $N = 899$ in the source recompute. They are excluded from corrected comparisons.

View	Accuracy	Macro-F1	κ
Summary	.2492	.1097	$-.0077$
Raw named features	.3426	.2453	.0066
Z-scored features	.1846	.0999	$-.0488$

S5.2 PATHOGAIT: Feature-SFT without Valid-Frame Fraction

The recovered generator removes the line named `valid_frame_fraction` from each feature-view record and asserts removal in every row. Remaining text, IDs, labels, and ordering are preserved. Its SFT data builder accepts `feature_vector_novff`. The source experiment log records the rerun and its training recipe, as well as a change of runtime environment. Full source input layers and training artifacts are external; the public predictions permit an independent check of the paired evaluation records and reported point estimates.

Table S37: PATHOGAIT feature-removal evaluation. Independently rescored generations on identical 2,181 test IDs and true labels; both have zero invalid outputs. The two test participants and single training realization limit inference.

Feature-SFT input	Accuracy	Macro-F1	κ
With valid-frame fraction	.6777	.6870	.6108
Without valid-frame fraction	.6410	.6316	.5649

The macro-F1 change is -0.0554 on matching test IDs. The ablated point estimate exceeds quality-free LR/SVM .564/.560, but two participants and one training realization preclude a population-level superiority claim. Runtime changes also prevent causal attribution of the whole difference to the field. The seven shared conditions exclude capture-quality features and use 1,440 PATHOGAIT clips; they contain no feature-SFT result.

S5.3 Release Scope and Source Rights

The correction bundle supplies split maps, checked generations, code snapshots, and verification code alongside the six-dataset scoring release. Full recordings, adapters, and complete shared-condition inputs remain external. Code/output licensing does not establish dataset or image rights; source credits list outstanding checks.

References

[1] Jacquelin Perry. *Gait Analysis: Normal and Pathological Function*. SLACK Incorporated, 1992.

[2] Richard Baker. Gait analysis methods in rehabilitation. *Journal of NeuroEngineering and Rehabilitation*, 3:4, 2006.

[3] Alberto Muro-de-la Herran, Begonya Garcia-Zapirain, and Amaia Mendez-Zorrilla. Gait analysis methods: An overview of wearable and non-wearable systems, highlighting clinical applications. *Sensors*, 14(2):3362–3394, 2014.

[4] Zhe Cao, Tomas Simon, Shih-En Wei, and Yaser Sheikh. Realtime multi-person 2D pose estimation using part affinity fields. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7291–7299, 2017.

[5] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. MediaPipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*, 2019.

[6] Valentin Bazarevsky, Ivan Grishchenko, Karthik Raveendran, Tyler Zhu, Fan Zhang, and Matthias Grundmann. BlazePose: On-device real-time body pose tracking. *arXiv preprint arXiv:2006.10204*, 2020.

[7] Jan Stenum, Kendra M. Cherry-Allen, Connor O. Pyles, Rachel D. Reetzke, Michael F. Vignos, and Ryan T. Roemmich. Applications of pose estimation in human health and performance across the lifespan. *Sensors*, 21(21):7315, 2021.

[8] Jan Stenum, Melody M. Hsu, Alexander Y. Pantelyat, and Ryan T. Roemmich. Clinical gait analysis using video-based pose estimation: Multiple perspectives, clinical populations, and measuring change. *PLOS Digital Health*, 3(3):e0000467, 2024.

[9] David A. Winter. *Biomechanics and Motor Control of Human Movement*. John Wiley & Sons, 4th edition, 2009.

[10] Michael H. Schwartz and Adam Rozumalski. The gait deviation index: A new comprehensive index of gait pathology. *Gait & Posture*, 28(3):351–357, 2008.

[11] Hisham Khalil, Ahmed Mohamed Saad Emam Saad, and Uswah Khairuddin. Diagnosis of cerebellar ataxia based on gait analysis using human pose estimation: A deep learning approach. In *2022 IEEE-EMBS Conference on Biomedical Engineering and Sciences (IECBES)*, pages 201–206, 2022.

[12] Daniel Deng, Jill L. Ostrem, Vy Nguyen, Daniel D. Cummins, Julia Sun, Anupam Pathak, Simon Little, and Reza Abbasi-Asl. Interpretable video-based tracking and quantification of parkinsonism clinical motor states. *npj Parkinson's Disease*, 10:122, 2024.

[13] Łukasz Kidziński, Bryan Yang, Jennifer L. Hicks, Apoorva Rajagopal, Scott L. Delp, and Michael H. Schwartz. Deep neural networks enable quantitative movement analysis using single-camera videos. *Nature Communications*, 11:4054, 2020.

[14] Mandy Lu, Kathleen L. Poston, Adolf Pfefferbaum, Edith V. Sullivan, Li Fei-Fei, Kilian M. Pohl, Juan Carlos Niebles, and Ehsan Adeli. Vision-based estimation of MDS-UPDRS gait scores for assessing parkinson's disease motor severity. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pages 637–647, 2020.

[15] Kooksung Jun, Yongwoo Lee, Sanghyub Lee, Deok-Won Lee, and Mun Sang Kim. Pathological gait classification using Kinect v2 and gated recurrent neural networks. *IEEE Access*, 8:139881–139891, 2020.

[16] Juri Opitz and Sebastian Burst. Macro F1 and Macro F1. *arXiv preprint arXiv:1911.03347*, 2019.

[17] Marina Sokolova and Guy Lapalme. A systematic analysis of performance measures for classification tasks. *Information Processing & Management*, 45(4):427–437, 2009.

[18] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D. Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 1877–1901, 2020.

[19] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

[20] Sheldon R. Simon. Quantification of human motion: Gait analysis—benefits and limitations to its application to clinical problems. *Journal of Biomechanics*, 37(12):1869–1880, 2004.

[21] Alexander Mathis, Pranav Mamidanna, Kevin M. Cury, Taiga Abe, Venkatesh N. Murthy, Mackenzie Weygandt Mathis, and Matthias Bethge. DeepLabCut: Markerless pose estimation of user-defined body parts with deep learning. *Nature Neuroscience*, 21(9):1281–1289, 2018.

[22] Scott D. Uhlrich, Antoine Falisse, Łukasz Kidziński, Julie Muccini, Michael Ko, Akshay S. Chaudhari, Jennifer L. Hicks, and Scott L. Delp. OpenCap: Human movement dynamics from smartphone videos. *PLOS Computational Biology*, 19(10): e1011462, 2023.

[23] Tanmay T. Verlekar, Henri De Vroey, Kurt Claeys, Hans Hallez, Luís D. Soares, and Paulo L. Correia. Estimation and validation of temporal gait features using a markerless 2D video system. *Computer Methods and Programs in Biomedicine*, 175:45–51, 2019.

[24] Masataka Yamamoto, Koji Shimatani, Masaki Hasegawa, Yuichi Kurita, Yuto Ishige, and Hiroshi Takemura. Accuracy of temporo-spatial and lower limb joint kinematics parameters using OpenPose for various gait patterns with orthosis. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 29:2666–2675, 2021.

[25] Andrea Sabo, Carolina Gorodetsky, Alfonso Fasano, Andrea Iaboni, and Babak Taati. Concurrent validity of Zeno instrumented walkway and video-based gait features in adults with parkinson's disease. *IEEE Journal of Translational Engineering in Health and Medicine*, 10:2100511, 2022.

[26] Sijie Yan, Yuanjun Xiong, and Dahua Lin. Spatial temporal graph convolutional networks for skeleton-based action recognition. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI)*, pages 7444–7452, 2018.

[27] Lei Shi, Yifan Zhang, Jian Cheng, and Hanqing Lu. Two-stream adaptive graph convolutional networks for skeleton-based action recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12026–12035, 2019.

[28] Ziyu Liu, Hongwen Zhang, Zhenghao Chen, Zhiyong Wang, and Wanli Ouyang. Disentangling and unifying graph convolutions for skeleton-based action recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 143–152, 2020.

[29] Elham Dolatabadi, Babak Taati, and Alex Mihailidis. An automated classification of pathological gait using unobtrusive sensing technology. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 25(12):2336–2346, 2017.

[30] Tanmay T. Verlekar, Luís D. Soares, and Paulo L. Correia. Automatic classification of gait impairments using a markerless 2D video-based system. *Sensors*, 18(9):2743, 2018.

[31] Javier Ortells, María Trinidad Herrero-Ezquerro, and Ramón A. Mollineda. Vision-based gait impairment analysis for aided diagnosis. *Medical & Biological Engineering & Computing*, 56(9):1553–1564, 2018.

[32] Jung Hwan Shin, Ri Yu, Jed Noel Ong, Chan Young Lee, Seung Ho Jeon, Hwanpil Park, Han Joon Kim, Jehee Lee, and Beomseok Jeon. Quantitative gait analysis using a pose-estimation algorithm with a single 2D-Video of Parkinson's Disease patients. *Journal of Parkinson's Disease*, 11(3):1271–1283, 2021.

[33] Andrea Sabo, Sina Mehdizadeh, Kimberley-Dale Ng, Andrea Iaboni, and Babak Taati. Assessment of Parkinsonian Gait in older adults with dementia via human pose tracking in video data. *Journal of NeuroEngineering and Rehabilitation*, 17:97, 2020.

[34] Qingyi Zeng, Peipei Liu, Ningbo Yu, Jialing Wu, Weiguang Huo, and Jianda Han. Video-based quantification of gait impairments in Parkinson's Disease using skeleton–silhouette fusion convolution network. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 31:2912–2922, 2023.

[35] Evelina Pantzar-Castilla, Andrea Cereatti, Giulio Figari, Nicolò Valeri, Gabriele Paolini, Ugo Della Croce, Anders Magnuson, and Jacques Riad. Knee joint sagittal plane movement in cerebral palsy: A comparative study of 2-dimensional markerless video and 3-dimensional gait analysis. *Acta Orthopaedica*, 89(6):656–661, 2018.

[36] Luca Lonini, Yaejin Moon, Kyle Embry, R. James Cotton, Kelly McKenzie, Sophia Jenz, and Arun Jayaraman. Video-based pose estimation for gait analysis in stroke survivors during clinical assessments: A proof-of-concept study. *Digital Biomarkers*, 6(1):9–18, 2022.

[37] Wasifur Rahman, Masum Hasan, Md Saiful Islam, Titilayo Olubajo, Jeet Thaker, Abdelrahman R. Abdelkader, Phillip Yang, Henry Paulson, Gulin Oz, Alexandra Durr, Thomas Klockgether, Tetsuo Ashizawa, READISCA Investigators, and Ehsan Hoque. Auto-Gait: Automatic ataxia risk assessment with computer vision from gait task videos. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 7(1):26:1–26:19, 2023.

[38] Katsuki Eguchi, Hiroaki Yaguchi, Hisashi Uwatoko, Yuki Iida, Shinsuke Hamada, Sanae Honma, Asako Takei, Fumio Moriwaka, and Ichiro Yabe. Feasibility of differentiating gait in Parkinson's Disease and spinocerebellar degeneration using a pose estimation algorithm in two-dimensional video. *Journal of the Neurological Sciences*, 464:123158, 2024.

[39] Rahm Ranjan, David Ahmedt-Aristizabal, Mohammad Ali Armin, and Juno Kim. Computer vision for clinical gait analysis: A gait abnormality video dataset. *arXiv preprint arXiv:2407.04190*, 2024.

[40] Tanmay T. Verlekar, Paulo L. Correia, and Luís D. Soares. Using transfer learning for classification of gait pathologies. In *2018 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pages 2376–2381, 2018.

[41] Pedro Albuquerque, Tanmay Tulsidas Verlekar, Paulo Lobato Correia, and Luís Ducla Soares. A spatiotemporal deep learning approach for automatic pathological gait classification. *Sensors*, 21(18):6202, 2021.

[42] Vida Adeli, Ivan Klabučar, Javad Rajabi, Benjamin Filtjens, Soroush Mehraban, Diwei Wang, Hyewon Seo, Trung-Hieu Hoang, Minh N. Do, Candice Muller, Claudia Neves de Oliveira, Daniel Boari Coelho, Pieter Ginis, Moran Gilat, Alice Nieuwboer, et al. CARE-PD: A multi-site anonymized clinical dataset for Parkinson's Disease gait assessment. In *Advances in Neural Information Processing Systems*, 2025. Datasets and Benchmarks Track.

[43] Biao Jiang, Xin Chen, Wen Liu, Jingyi Yu, Gang Yu, and Tao Chen. MotionGPT: Human motion as a foreign language. In *Advances in Neural Information Processing Systems*, pages 20067–20079, 2023.

[44] Ling-Hao Chen, Shunlin Lu, Ailing Zeng, Hao Zhang, Benyou Wang, Ruimao Zhang, and Lei Zhang. MotionLLM: Understanding human behaviors from human motions and videos. *arXiv preprint arXiv:2405.20340*, 2024.

[45] Jiawei Mo, Yixuan Chen, Rifen Lin, Yongkang Ni, Min Zeng, Xiping Hu, and Min Li. MoChat: Joints-grouped spatio-temporal grounding LLM for multi-turn motion comprehension and description. *arXiv preprint arXiv:2410.11404*, 2024.

[46] Shaopeng Yang, Jilong Wang, Saihui Hou, Xu Liu, Chunshui Cao, Liang Wang, and Yongzhen Huang. Bridging gait recognition and large language models sequence modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3460–3469, 2025.

[47] Diwei Wang, Cédric Bobenrieth, and Hyewon Seo. AGIR: Assessing 3D gait impairment with reasoning based on LLMs. *arXiv preprint arXiv:2503.18141*, 2025.

[48] Ruize Yang, Ann Kennedy, and R. James Cotton. BiomechGPT: Towards a biomechanically fluent multimodal foundation model for clinically relevant motion tasks. *arXiv preprint arXiv:2505.18465v1*, 2025.

[49] Erdong Chen, Yuyang Ji, Jacob K. Greenberg, Benjamin Steel, Faraz Arkam, Abigail Lewis, Pranay Singh, and Feng Liu. BioGait-VLM: A tri-modal vision-language-biomechanics framework for interpretable clinical gait assessment. *arXiv preprint arXiv:2603.08564*, 2026.

[50] Carlo Dindorf, Jonas Dully, Rebecca Keilhauer, Michael Lorenz, and Michael Fröhlich. Evaluating large language models for gait classification using text-encoded kinematic waveforms. *arXiv preprint arXiv:2603.13317*, 2026.

[51] Xi Fang, Weijie Xu, Fiona Anting Tan, Jiani Zhang, Ziqing Hu, Yanjun Qi, Scott Nickleach, Diego Socolinsky, Srinivasan Sengamedu, and Christos Faloutsos. Large language models (LLMs) on tabular data: Prediction, generation, and understanding—a survey. *Transactions on Machine Learning Research*, 2024.

[52] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems*, pages 34892–34916, 2023.

[53] Shukang Yin, Chaoyou Fu, Sirui Zhao, Ke Li, Xing Sun, Tong Xu, and Enhong Chen. A survey on multimodal large language models. *National Science Review*, 11(12):nwae403, 2024.

[54] Catalin Ionescu, Dragos Papava, Vlad Olaru, and Cristian Sminchisescu. Human3.6M: Large scale datasets and predictive methods for 3D human sensing in natural environments. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 36(7):1325–1339, 2014.

[55] Robert Palisano, Peter Rosenbaum, Stephen Walter, Dianne Russell, Ellen Wood, and Barbara Galuppi. Development and reliability of a system to classify gross motor function in children with cerebral palsy. *Developmental Medicine & Child Neurology*, 39(4):214–223, 1997.

[56] Hisham Khalil, Ahmed Saad, and Uswah Khairuddin. Dataset for gait analysis of cerebellar ataxic patients and healthy adults using MediaPipe pose. Mendeley Data, 2023. Version 1.

[57] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

[58] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

[59] Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, et al. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025.

[60] Team GLM, Aohan Zeng, Bin Xu, Bowen Wang, Chenhui Zhang, Da Yin, Dan Zhang, Diego Rojas, Guanyu Feng, Hanlin Zhao, et al. ChatGLM: A family of large language models from GLM-130B to GLM-4 all tools. *arXiv preprint arXiv:2406.12793*, 2024.

[61] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645:633–638, 2025.

[62] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 24824–24837, 2022.

[63] Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine Learning*, 20(3):273–297, 1995.

[64] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[65] Jacob Cohen. Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit. *Psychological Bulletin*, 70(4):213–220, 1968.

[66] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993.

[67] Ahmed A. Al-Jubouri, Israa Hadi Ali, and Yasen Rajihy. Gait and full body movement dataset of autistic children classified by rough set classifier. *Journal of Physics: Conference Series*, 1818(1):012201, 2021.